

Panduan Penggunaan Statify

Modul Analisis *Time Series*

Pengantar

Panduan ini dibuat sebagai dokumentasi penggunaan aplikasi Statify modul analisis *time series* yang terbagi atas beberapa menu antara lain, menu *smoothing*, *decomposition*, *autocorrelation*, *unit root test*, dan *box-jenkins model*. Adapun pada bagian selanjutnya akan dijelaskan mengenai rincian panduan penggunaan dari masing-masing menu analisis *time series*.

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Proses Input Data

1. Tahapan Input Data

Ikuti langkah-langkah berikut.

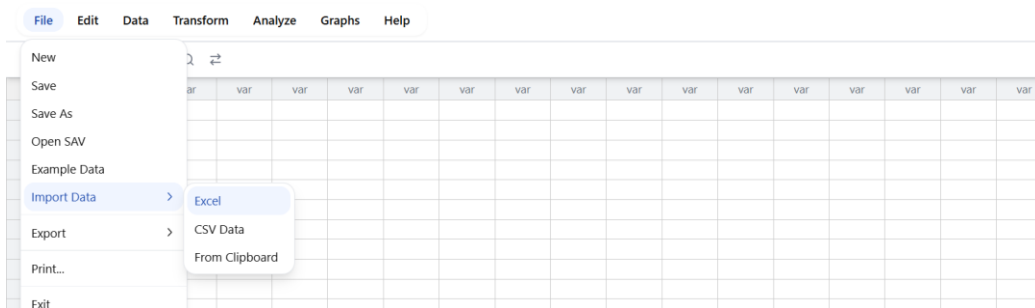
1. Unduh data berikut

[Data Penumpang.xlsx](#).

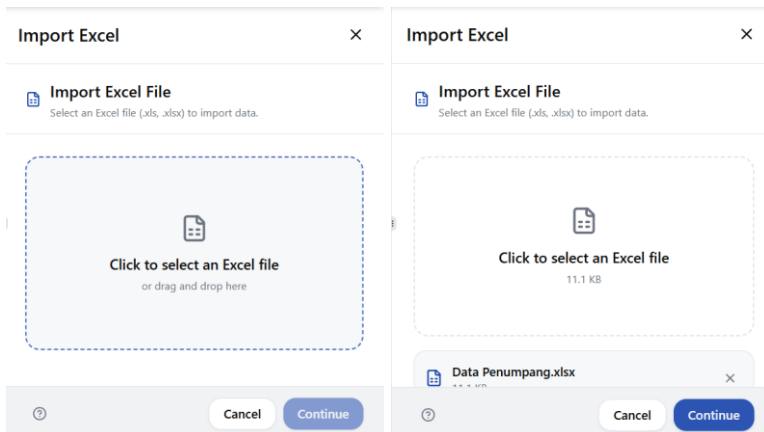
2. Buka situs web Statify

<https://statify-dev.student.stis.ac.id/dashboard/data>.

3. Klik File → Import Data → Excel pada navbar Statify

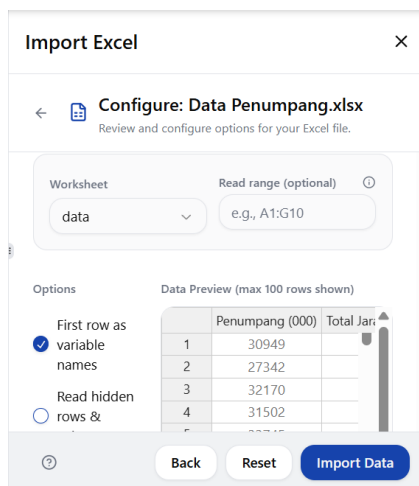


4. Klik untuk memilih file excel yang hendak diimpor



5. Klik continue untuk melanjutkan.

6. Gunakan konfigurasi *default*-nya saja lalu klik Import Data.



Import Excel

Configure: Data Penumpang.xlsx
Review and configure options for your Excel file.

Worksheet: data
Read range (optional): e.g., A1:G10

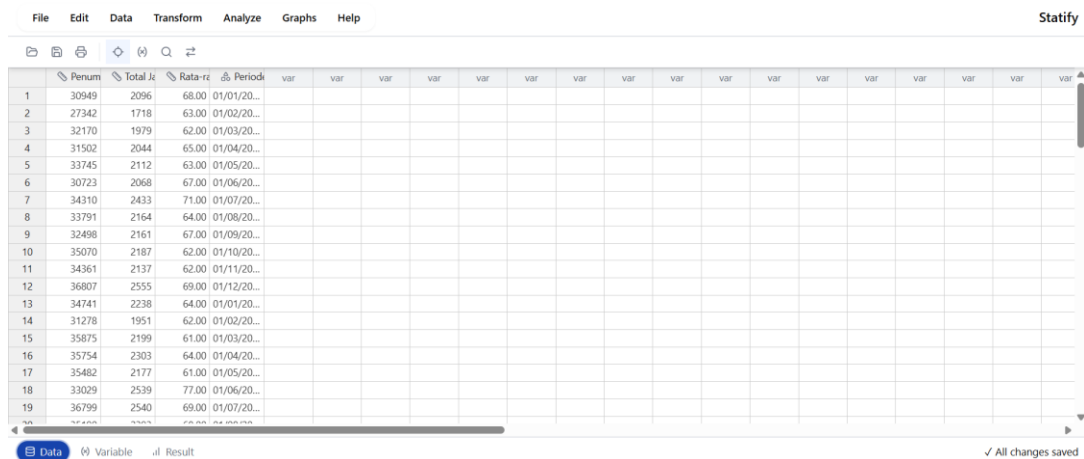
Options:
First row as variable names: ☒
Read hidden rows & columns: ☐

Data Preview (max 100 rows shown):

	Penumpang (000)	Total Jari
1	30949	
2	27342	
3	32170	
4	31502	
5	33745	
6	30723	
7	34310	
8	33791	
9	32498	
10	35070	
11	34361	
12	36807	
13	34741	
14	31278	
15	35875	
16	35754	
17	35482	
18	33029	
19	36799	

Back Reset Import Data

7. Pastikan data yang masuk sudah diimpor dengan benar.



File Edit Data Transform Analyze Graphs Help

Statify

	Penum	Total Ji	Rata-r	Periode	var	var	var	var	var	var	var	var	var	var	var	var	var	var	var
1	30949	2096	68.00	01/01/20...															
2	27342	1718	63.00	01/02/20...															
3	32170	1979	62.00	01/03/20...															
4	31502	2044	65.00	01/04/20...															
5	33745	2112	63.00	01/05/20...															
6	30723	2068	67.00	01/06/20...															
7	34310	2433	71.00	01/07/20...															
8	33791	2164	64.00	01/08/20...															
9	32498	2161	67.00	01/09/20...															
10	35070	2187	62.00	01/10/20...															
11	34361	2137	62.00	01/11/20...															
12	36807	2555	69.00	01/12/20...															
13	34741	2238	64.00	01/01/20...															
14	31278	1951	62.00	01/02/20...															
15	35875	2199	61.00	01/03/20...															
16	35754	2303	64.00	01/04/20...															
17	35482	2177	61.00	01/05/20...															
18	33029	2539	77.00	01/06/20...															
19	36799	2540	69.00	01/07/20...															

Data Variable Result

✓ All changes saved

8. Data sudah siap digunakan untuk melakukan analisis *time series* pada tahapan-tahapan berikutnya.

Keterangan Data

Data yang digunakan adalah data penumpang kereta api nasional bulanan periode Januari 2017 hingga Desember 2024 dengan jumlah sembilan puluh enam observasi yang terdiri atas empat variabel sebagai berikut.

1. Penumpang (000) : Total penumpang kereta api nasional dalam satuan ribu.
2. TJTP (000.000 km) : Total jarak tempuh penumpang dalam satuan ratusan ribu kilometer.
3. RJP (km) : Rata-rata jarak perjalanan per penumpang dalam satuan kilometer.

4. Periode : Periode data per bulan.

Data bersumber dari *website* resmi Badan Pusat Statistik (BPS) dari *link* berikut :

<https://www.bps.go.id/id/statistics-table/2/MjM1MyMy/penumpang-angkutan-kereta-api-nasional-bulanan-.html>.

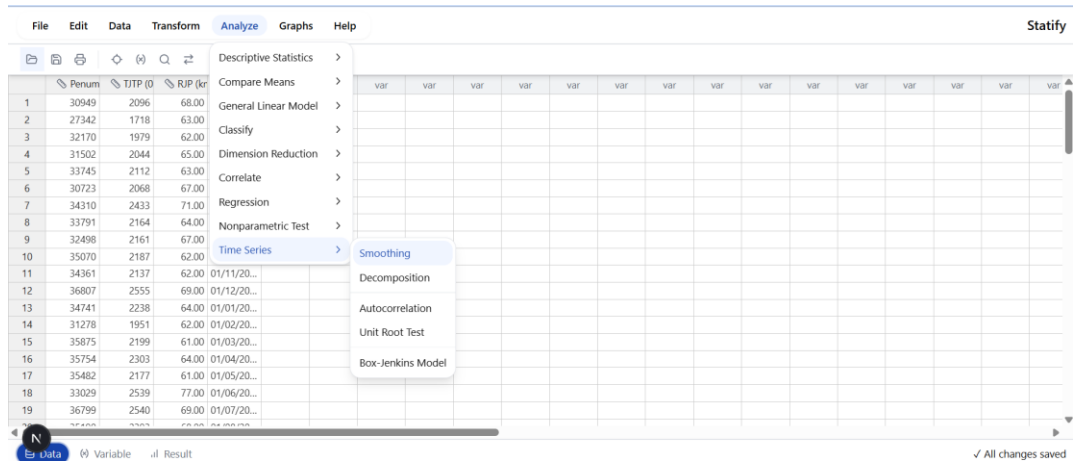
Smoothing

2. Tahapan Menggunakan Menu *Smoothing*

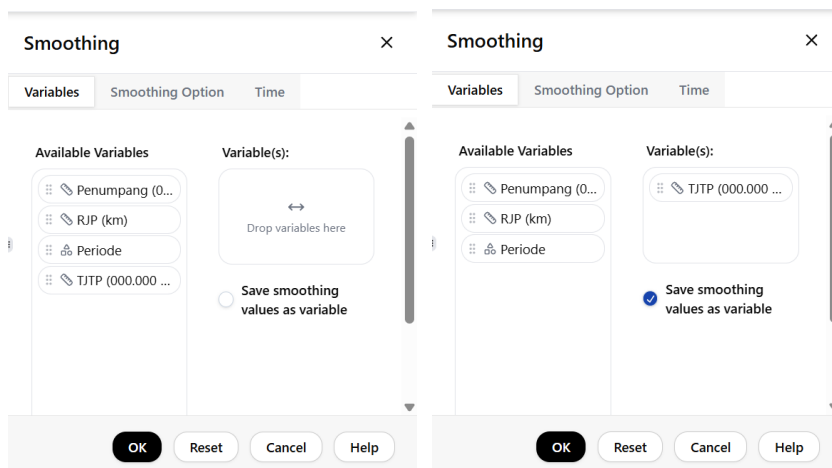
2.1. *Simple Moving Average*

Ikuti langkah-langkah berikut:

1. Klik Analyze → Time Series → Smoothing untuk masuk ke menu *Smoothing*.



2. Pilih variabel yang ingin digunakan dan klik save smoothing values as variable untuk menyimpan hasil *smoothing* sebagai variabel baru.



3. Pilih tab Smoothing Option untuk memilih metode *smoothing*.

The screenshot shows the 'Smoothing' dialog box with the 'Smoothing Option' tab selected. The 'Smoothing Method' section contains a 'method' dropdown menu set to 'Simple Moving Average' and a 'distance' input field set to '2'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

4. Atur parameter metode Simple Moving Average yang diinginkan.

This screenshot is similar to the previous one, but the 'distance' input field is now set to '5'. Below the dialog box, a status message reads '✓ All changes saved'.

5. Pilih tab time untuk menyesuaikan periode waktu.

The screenshot shows the 'Smoothing' dialog box with the 'Time' tab selected. The 'Time Option' section contains a 'time spesification' dropdown menu set to 'Not Dated', a 'periodicity' label with the text 'don't have periodicity', and a 'start time' label with the text '00:00'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

6. Sesuaikan time specification dengan periode data seperti konfigurasi berikut.

Smoothing

Variables

Smoothing Option

Time

Time Option

time specification : Years-Months

year : 2017

month : 1

periodicity : 12

start time : Jan 2017

OK

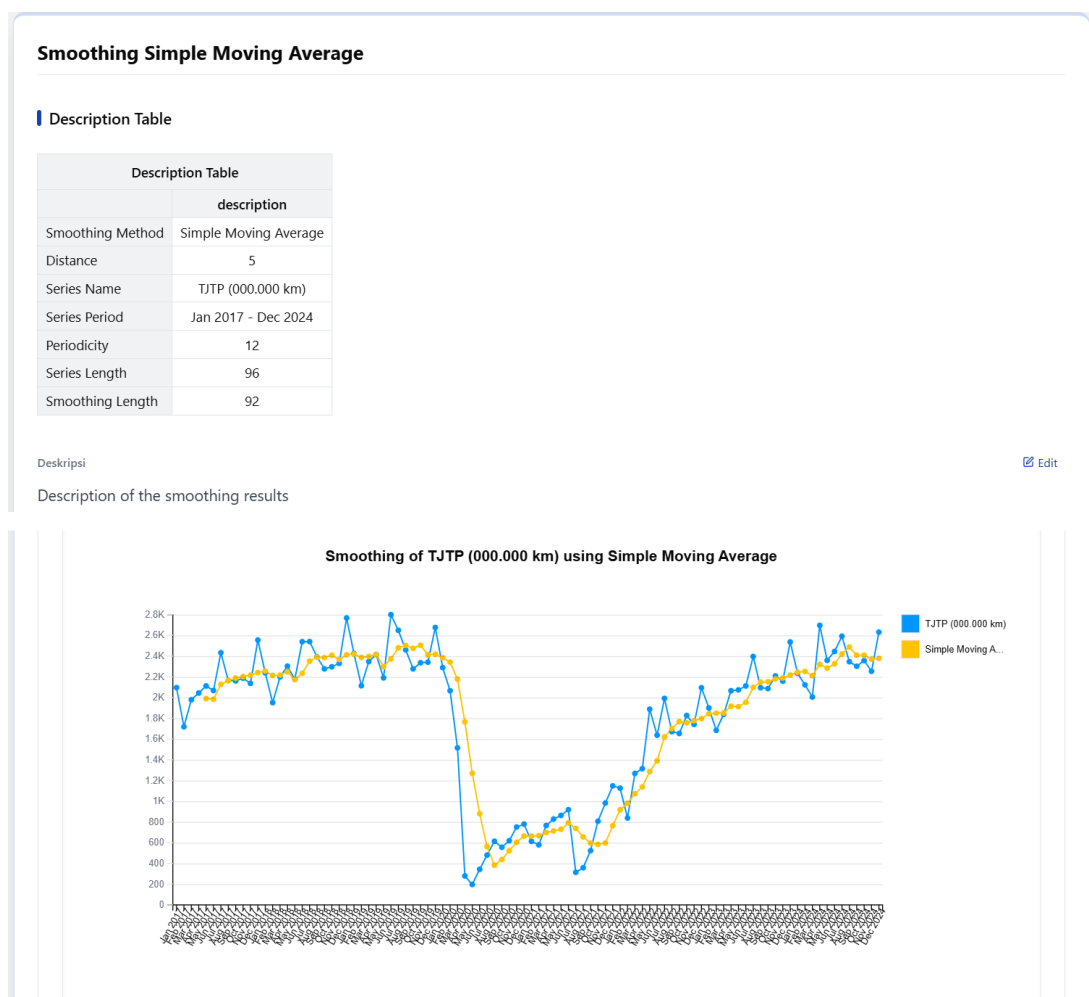
Reset

Cancel

Help

7. Setelah itu klik ok.

8. Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.



Smoothing Evaluation	
Smoothing Evaluation Results	
	value
MSE	81048.968
RMSE	284.691
MAE	187.280
MPE	-12.357
MAPE	25.047

Deskripsi

Smoothing evaluation results

[Edit](#)

- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

File Edit Data Transform Analyze Graphs Help																Statify	
	Penum	TJTP (0	RJP (kr	Period	TJTP_000.000_kn_sma_5	var	var	var	var	var	var	var	var	var	var	var	var
1	30949	2096	68.00	01/01/20...	NaN												
2	27342	1718	63.00	01/02/20...	NaN												
3	32170	1979	62.00	01/03/20...	NaN												
4	31502	2044	65.00	01/04/20...	NaN												
5	33745	2112	63.00	01/05/20...	1989.80												
6	30723	2068	67.00	01/06/20...	1984.20												
7	34310	2433	71.00	01/07/20...	2127.20												
8	33791	2164	64.00	01/08/20...	2164.20												
9	32498	2161	67.00	01/09/20...	2187.60												
10	35070	2187	62.00	01/10/20...	2202.60												
11	34361	2137	62.00	01/11/20...	2216.40												
12	36807	2555	69.00	01/12/20...	2240.80												
13	34741	2238	64.00	01/01/20...	2255.60												
14	31278	1951	62.00	01/02/20...	2213.60												
15	35875	2199	61.00	01/03/20...	2216.00												
16	35754	2303	64.00	01/04/20...	2249.20												
17	35482	2177	61.00	01/05/20...	2173.60												
18	33029	2539	77.00	01/06/20...	2233.80												
19	36799	2540	69.00	01/07/20...	2351.60												

Statify

✓ All changes saved

2.2. Double Moving Average

Ikuti langkah-langkah berikut:

- Klik Analyze → Time Series → Smoothing untuk masuk ke menu *Smoothing*.

File Edit Data Transform Analyze Graphs Help																Statify	
	Penum	TJTP (0	RJP (kr	Period	TJTP_000.000_kn_sma_5	var	var	var	var	var	var	var	var	var	var	var	var
1	30949	2096	68.00	01/01/20...	NaN												
2	27342	1718	63.00	01/02/20...	NaN												
3	32170	1979	62.00	01/03/20...	NaN												
4	31502	2044	65.00	01/04/20...	NaN												
5	33745	2112	63.00	01/05/20...	1989.80												
6	30723	2068	67.00	01/06/20...	1984.20												
7	34310	2433	71.00	01/07/20...	2127.20												
8	33791	2164	64.00	01/08/20...	2164.20												
9	32498	2161	67.00	01/09/20...	2187.60												
10	35070	2187	62.00	01/10/20...	2202.60												
11	34361	2137	62.00	01/11/20...	2216.40												
12	36807	2555	69.00	01/12/20...	2240.80												
13	34741	2238	64.00	01/01/20...	2255.60												
14	31278	1951	62.00	01/02/20...	2213.60												
15	35875	2199	61.00	01/03/20...	2216.00												
16	35754	2303	64.00	01/04/20...	2249.20												
17	35482	2177	61.00	01/05/20...	2173.60												
18	33029	2539	77.00	01/06/20...	2233.80												
19	36799	2540	69.00	01/07/20...	2351.60												

Statify

✓ All changes saved

2. Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.

The screenshot shows the 'Smoothing' dialog box with the 'Variables' tab selected. On the left, under 'Available Variables', there are four items: 'Penumpang (0...', 'RJP (km)', 'Periode', and 'TJTP (000.000 ...'. On the right, under 'Variable(s):', the 'TJTP (000.000 ...' variable is selected. Below this, there is a checked checkbox labeled 'Save smoothing values as variable'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

3. Pilih tab Smoothing Option untuk memilih metode *smoothing* lalu pilih metode Double Moving Average dan atur parameter yang diinginkan.

The screenshot shows the 'Smoothing' dialog box with the 'Smoothing Option' tab selected. Under the 'Smoothing Method' section, the 'method:' dropdown is set to 'Double Moving Average' and the 'distance:' spinner is set to '3'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

4. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya. Sebagai catatan konfigurasi waktu untuk menu time series cukup dilakukan sekali saja karena sudah terintegrasi untuk setiap menu *time series*.

The screenshot shows the 'Smoothing' dialog box with the 'Time' tab selected. Under the 'Time Option' section, the 'time spesification:' dropdown is set to 'Years-Months', the 'year:' is '2017', the 'month:' is '1', the 'periodicity:' is '12', and the 'start time:' is 'Jan 2017'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.



- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

The screenshot shows the Statify software interface. At the top, there is a menu bar with 'File', 'Edit', 'Data', 'Transform', 'Analyze', 'Graphs', and 'Help'. The main area displays a data table with columns for variables: 'Penum', 'TJTP (0', 'RJP (kr', 'TJTP_000.000_km_sma_5', 'TJTP_000.000_km_dma_3', and several 'var' columns. The data rows contain numerical values and dates. At the bottom left, there is a 'Data' button with a blue icon. At the bottom right, there is a status bar that says 'All changes saved'.

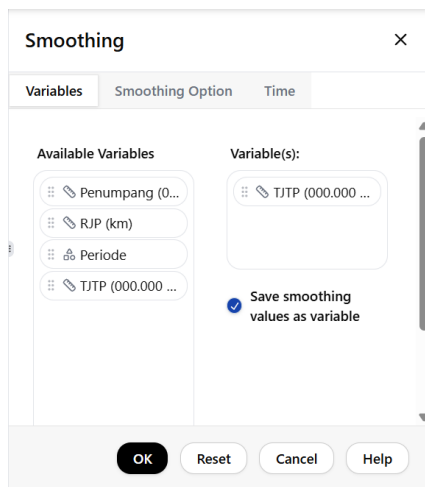
2.3. Simple Exponential Smoothing

Ikuti langkah-langkah berikut:

- Klik *Analyze* → *Time Series* → *Smoothing* untuk masuk ke menu *Smoothing*.

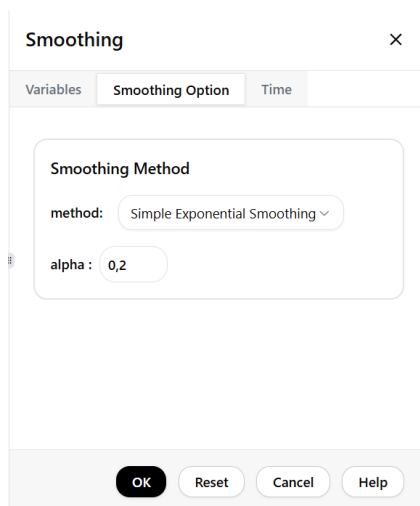
The screenshot shows the Statify software interface with the 'Analyze' menu open. The menu options are: 'Descriptive Statistics', 'Compare Means', 'General Linear Model', 'Classify', 'Dimension Reduction', 'Correlate', 'Regression', 'Nonparametric Test', 'Time Series', and 'Smoothing'. The 'Time Series' option is highlighted, and a sub-menu is visible with options: 'Smoothing', 'Decomposition', 'Autocorrelation', 'Unit Root Test', and 'Box-Jenkins Model'. The 'Smoothing' option is highlighted in the sub-menu. The background shows the same data table as in the previous screenshot.

2. Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.



The screenshot shows the 'Smoothing' dialog box with the 'Variables' tab selected. The 'Available Variables' list on the left includes 'Penumpang (0...', 'RJP (km)', 'Periode', and 'TJTP (000.000 ...'. The 'Variable(s):' list on the right contains 'TJTP (000.000 ...'. A checkbox labeled 'Save smoothing values as variable' is checked. At the bottom, there are buttons for 'OK', 'Reset', 'Cancel', and 'Help'.

3. Pilih tab Smoothing Option untuk memilih metode *smoothing* lalu pilih metode Simple Exponential Smoothing dan atur parameter yang diinginkan.



The screenshot shows the 'Smoothing' dialog box with the 'Smoothing Option' tab selected. The 'Smoothing Method' section contains a 'method:' dropdown menu set to 'Simple Exponential Smoothing' and an 'alpha:' input field set to '0,2'. At the bottom, there are buttons for 'OK', 'Reset', 'Cancel', and 'Help'.

4. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan

karena aplikasi menyimpan hasil konfigurasi sebelumnya.

Smoothing

Variables

Smoothing Option

Time

Time Option

time spesification : Years-Months

year : 2017

month : 1

periodicity : 12

start time : Jan 2017

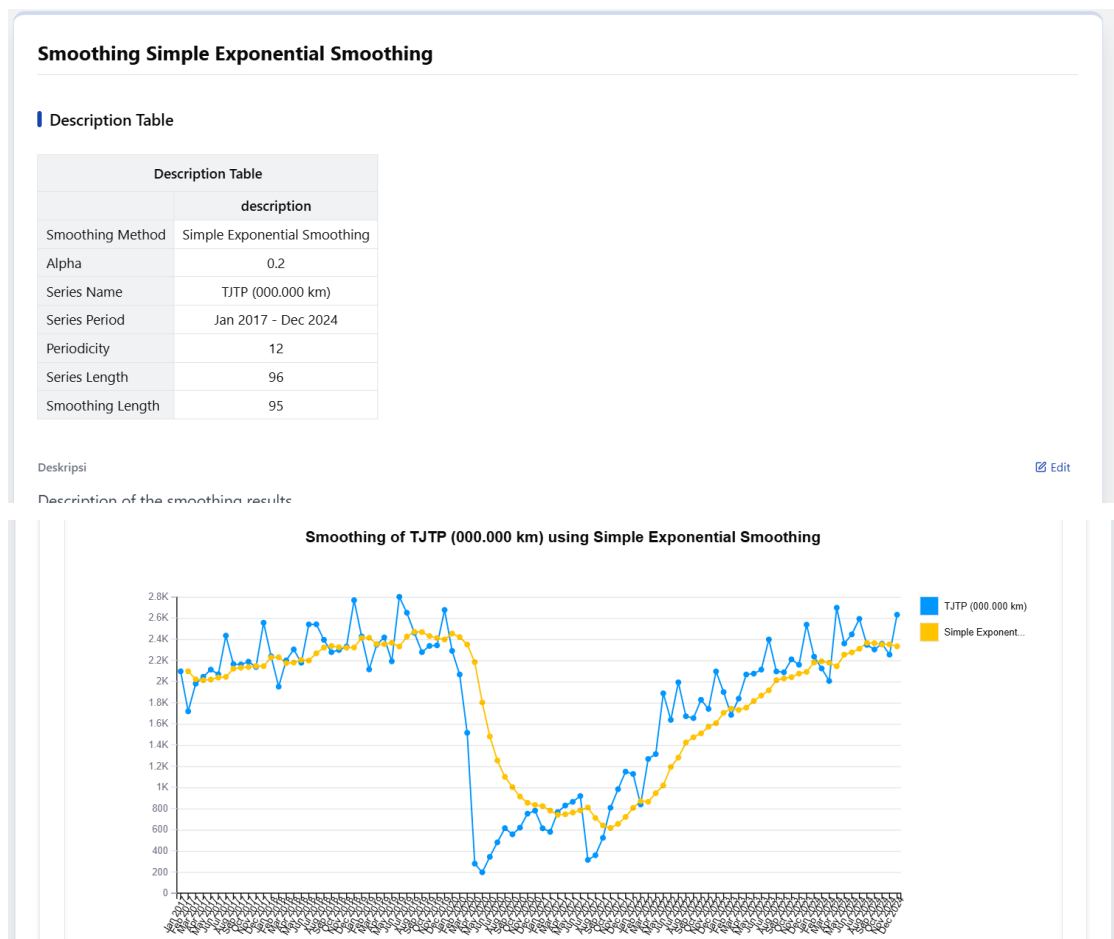
OK

Reset

Cancel

Help

- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.



Smoothing Evaluation	
Smoothing Evaluation Results	
	value
MSE	162606.919
RMSE	403.245
MAE	261.733
MPE	-21.481
MAPE	36.454

Deskripsi

Smoothing evaluation results

[Edit](#)

- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

File Edit Data Transform Analyze Graphs Help														Statify
	Penum	TJTP (0	RJP (kr	Period	TJTP_000.000_km_sma_5	TJTP_000.000_km_dma_3	TJTP_000.000_km_ses_0.2	var	var	var	var	var	var	var
1	30949	2096	68.00	01/01/20...	NaN	NaN	NaN							
2	27342	1718	63.00	01/02/20...	NaN	NaN	2096.00							
3	32170	1979	62.00	01/03/20...	NaN	NaN	2020.40							
4	31502	2044	65.00	01/04/20...	NaN	NaN	2012.12							
5	33745	2112	63.00	01/05/20...	1989.80	2208.56	2018.50							
6	30723	2068	67.00	01/06/20...	1984.20	2201.78	2037.20							
7	34310	2433	71.00	01/07/20...	2127.20	2397.00	2043.36							
8	33791	2164	64.00	01/08/20...	2164.20	2331.22	2121.29							
9	32498	2161	67.00	01/09/20...	2187.60	2305.56	2129.83							
10	35070	2187	62.00	01/10/20...	2202.60	2082.00	2136.06							
11	34361	2137	62.00	01/11/20...	2216.40	2095.00	2146.25							
12	36807	2555	69.00	01/12/20...	2240.80	2462.11	2144.40							
13	34741	2238	64.00	01/01/20...	2255.60	2420.22	2226.52							
14	31278	1951	62.00	01/02/20...	2213.60	2176.67	2228.82							
15	35875	2199	61.00	01/03/20...	2216.00	1929.78	2173.25							
16	35754	2303	64.00	01/04/20...	2249.20	2100.78	2178.40							
17	35482	2177	61.00	01/05/20...	2173.60	2341.22	2203.32							
18	33029	2539	77.00	01/06/20...	2233.80	2541.00	2198.06							
19	36799	2540	69.00	01/07/20...	2351.60	2599.56	2266.25							

[Data](#)
[Variable](#)
[Result](#)

✓ All changes saved

2.4. Double Exponential Smoothing

Ikuti langkah-langkah berikut:

- Klik Analyze → Time Series → Smoothing untuk masuk ke menu *Smoothing*.

File Edit Data Transform Analyze Graphs Help														Statify
	Penum	TJTP (0	RJP (kr	Period	TJTP_000.000_km_sma_5	TJTP_000.000_km_dma_3	TJTP_000.000_km_ses_0.2	var	var	var	var	var	var	var
1	30949	2096	68.00	01/01/20...	NaN	NaN	NaN							
2	27342	1718	63.00	01/02/20...	NaN	NaN	2096.00							
3	32170	1979	62.00	01/03/20...	NaN	NaN	2020.40							
4	31502	2044	65.00	01/04/20...	NaN	NaN	2012.12							
5	33745	2112	63.00	01/05/20...	1989.80	2208.56	2018.50							
6	30723	2068	67.00	01/06/20...	1984.20	2201.78	2037.20							
7	34310	2433	71.00	01/07/20...	2127.20	2397.00	2043.36							
8	33791	2164	64.00	01/08/20...	2164.20	2331.22	2121.29							
9	32498	2161	67.00	01/09/20...	2187.60	2305.56	2129.83							
10	35070	2187	62.00	01/10/20...	2202.60	2082.00	2136.06							
11	34361	2137	62.00	01/11/20...	2216.40	2095.00	2146.25							
12	36807	2555	69.00	01/12/20...	2240.80	2462.11	2144.40							
13	34741	2238	64.00	01/01/20...	2255.60	2420.22	2226.52							
14	31278	1951	62.00	01/02/20...	2213.60	2176.67	2228.82							
15	35875	2199	61.00	01/03/20...	2216.00	1929.78	2173.25							
16	35754	2303	64.00	01/04/20...	2249.20	2100.78	2178.40							
17	35482	2177	61.00	01/05/20...	2173.60	2341.22	2203.32							
18	33029	2539	77.00	01/06/20...	2233.80	2541.00	2198.06							
19	36799	2540	69.00	01/07/20...	2351.60	2599.56	2266.25							

[Data](#)
[Variable](#)
[Result](#)

✓ All changes saved

2. Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.

The screenshot shows the 'Smoothing' dialog box with the 'Variables' tab selected. On the left, under 'Available Variables', there is a list: 'Penumpang (0...', 'RJP (km)', 'Periode', and three instances of 'TJTP (000.000 ...'. On the right, under 'Variable(s):', the first instance of 'TJTP (000.000 ...' is selected. Below this list, there is a checkbox labeled 'Save smoothing values as variable' which is checked. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

3. Pilih tab Smoothing Option untuk memilih metode *smoothing* lalu pilih metode Double Exponential Smoothing dan atur parameter yang diinginkan.

The screenshot shows the 'Smoothing' dialog box with the 'Smoothing Option' tab selected. A 'Smoothing Method' section contains a 'method:' dropdown menu set to 'Double Exponential Smoothing' and an 'alpha:' input field with the value '0,4'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

4. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya.

The screenshot shows the 'Smoothing' dialog box with the 'Time' tab selected. A 'Time Option' section contains a 'time spesification:' dropdown menu set to 'Years-Months', a 'year:' input field with '2017', a 'month:' input field with '1', a 'periodicity:' label with '12', and a 'start time:' label with 'Jan 2017'. At the bottom, there are four buttons: 'OK', 'Reset', 'Cancel', and 'Help'.

- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.



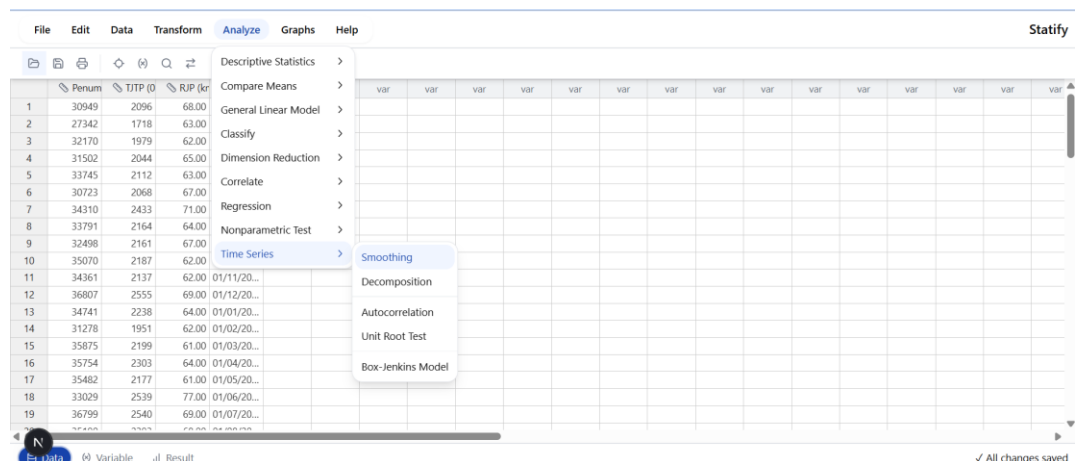
- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

The screenshot shows a software interface with a menu bar (File, Edit, Data, Transform, Analyze, Graphs, Help) and a toolbar. Below the toolbar is a data table with columns: Penumpang, TJTP (km), RJP (km), Periode, TJTP_000.000_km_sma_5, TJTP_000.000_km_dma_3, TJTP_000.000_km_ses_0.2, TJTP_000.000_km_des_0.4, and several 'var' columns. The 'Data' button is highlighted at the bottom left. A status bar at the bottom right indicates 'All changes saved'.

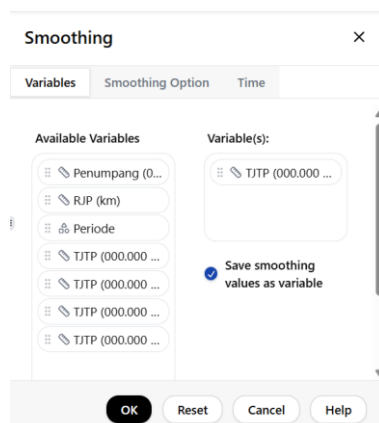
2.5. Holt's Method Exponential Smoothing

Ikuti langkah-langkah berikut:

- Klik Analyze → Time Series → Smoothing untuk masuk ke menu *Smoothing*.



- Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.



- Pilih tab Smoothing Option untuk memilih metode *smoothing* lalu pilih metode Holt's Exponential Smoothing dan atur parameter yang diinginkan.

The screenshot shows a dialog box titled "Smoothing" with a close button (X) in the top right corner. It has three tabs: "Variables", "Smoothing Option", and "Time". The "Smoothing Option" tab is selected. Inside the dialog, there is a section titled "Smoothing Method" with a dropdown menu labeled "method:" set to "Holt's Method Exponential Smooth". Below this are two input fields: "alpha:" with the value "0,1" and "beta:" with the value "0,4". At the bottom of the dialog are four buttons: "OK", "Reset", "Cancel", and "Help".

- Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya.

The screenshot shows the same "Smoothing" dialog box, but with the "Time" tab selected. The "Time Option" section contains a dropdown menu labeled "time spesification:" set to "Years-Months". Below this are four input fields: "year:" with the value "2017", "month:" with the value "1", "periodicity:" with the value "12", and "start time:" with the value "Jan 2017". The "OK", "Reset", "Cancel", and "Help" buttons are still present at the bottom.

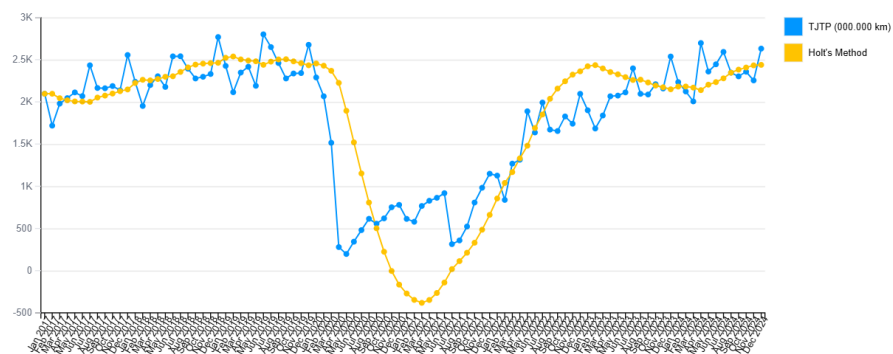
- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.

Smoothing Holt's Method Exponential Smoothing

Description Table

Description Table	
	description
Smoothing Method	Holt's Method
Alpha	0.1
Beta	0.4
Series Name	TJTP (000.000 km)
Series Period	Jan 2017 - Dec 2024
Periodicity	12
Series Length	96
Smoothing Length	96

Smoothing of TJTP (000.000 km) using Holt's Method



Smoothing Evaluation

Smoothing Evaluation Results	
	value
MSE	246621.139
RMSE	496.610
MAE	338.060
MPE	-8.883
MAPE	45.701

Deskripsi

Smoothing evaluation results

[Edit](#)

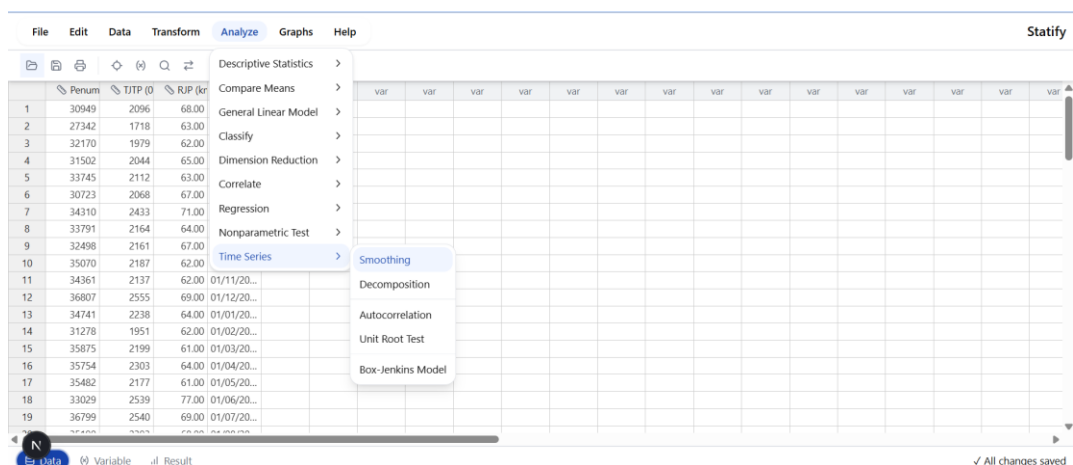
- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

	Penumpang	TJTP (0	RJP (kr	Periode	TJTP_000.000_km_sma_5	TJTP_000.000_km_dma_3	TJTP_000.000_km_ses_0.2	TJTP_000.000_km_des_0.4	TJTP_000.000_km_holt_0.1_0.4	var
1	30949	2096	68.00	01/01/20...	NaN	NaN	NaN	NaN	2096.00	
2	27342	1718	63.00	01/02/20...	NaN	NaN	2096.00	NaN	2096.00	
3	32170	1979	62.00	01/03/20...	NaN	NaN	2020.40	1692.80	2043.08	
4	31502	2044	65.00	01/04/20...	NaN	NaN	2012.12	1830.08	2018.99	
5	33745	2112	63.00	01/05/20...	1989.80	2208.56	2018.50	1972.66	2004.81	
6	30723	2068	67.00	01/06/20...	1984.20	2201.78	2037.20	2107.94	2003.13	
7	34310	2433	71.00	01/07/20...	2127.20	2397.00	2043.36	2110.35	1999.82	
8	33791	2164	64.00	01/08/20...	2164.20	2331.22	2121.29	2493.78	2050.66	
9	32498	2161	67.00	01/09/20...	2187.60	2305.56	2129.83	2335.16	2074.06	
10	35070	2187	62.00	01/10/20...	2202.60	2082.00	2136.06	2247.47	2098.29	
11	34361	2137	62.00	01/11/20...	2216.40	2095.00	2146.25	2229.80	2126.25	
12	36807	2555	69.00	01/12/20...	2240.80	2462.11	2144.40	2163.26	2146.84	
13	34741	2238	64.00	01/01/20...	2255.60	2420.22	2226.52	2580.97	2223.50	
14	31278	1951	62.00	01/02/20...	2213.60	2176.67	2228.82	2389.05	2261.37	
15	35875	2199	61.00	01/03/20...	2216.00	1929.78	2173.25	1989.66	2254.34	
16	35754	2303	64.00	01/04/20...	2249.20	2100.78	2178.40	2104.23	2270.60	
17	35482	2177	61.00	01/05/20...	2173.60	2341.22	2203.32	2271.57	2296.93	
18	33029	2539	77.00	01/06/20...	2233.80	2541.00	2198.06	2202.44	2303.23	
19	36799	2540	69.00	01/07/20...	2351.60	2599.56	2266.25	2559.62	2354.53	

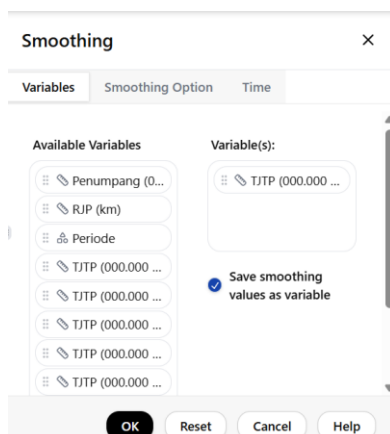
2.6. Winter's Method Exponential Smoothing

Ikuti langkah-langkah berikut:

- Klik Analyze → Time Series → Smoothing untuk masuk ke menu *Smoothing*.



- Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.



- Pilih tab Smoothing Option untuk memilih metode *smoothing* lalu pilih metode Winter's Exponential Smoothing dan atur parameter yang diinginkan

The screenshot shows the 'Smoothing' dialog box with the 'Smoothing Option' tab selected. The 'Smoothing Method' is set to 'Winter's Method Exponential Smr'. The parameters are: alpha: 0,2, beta: 0,1, and gamma: 0,7. A note at the bottom states: 'note: winters method need time spesification with periodicity'. The 'OK' button is highlighted.

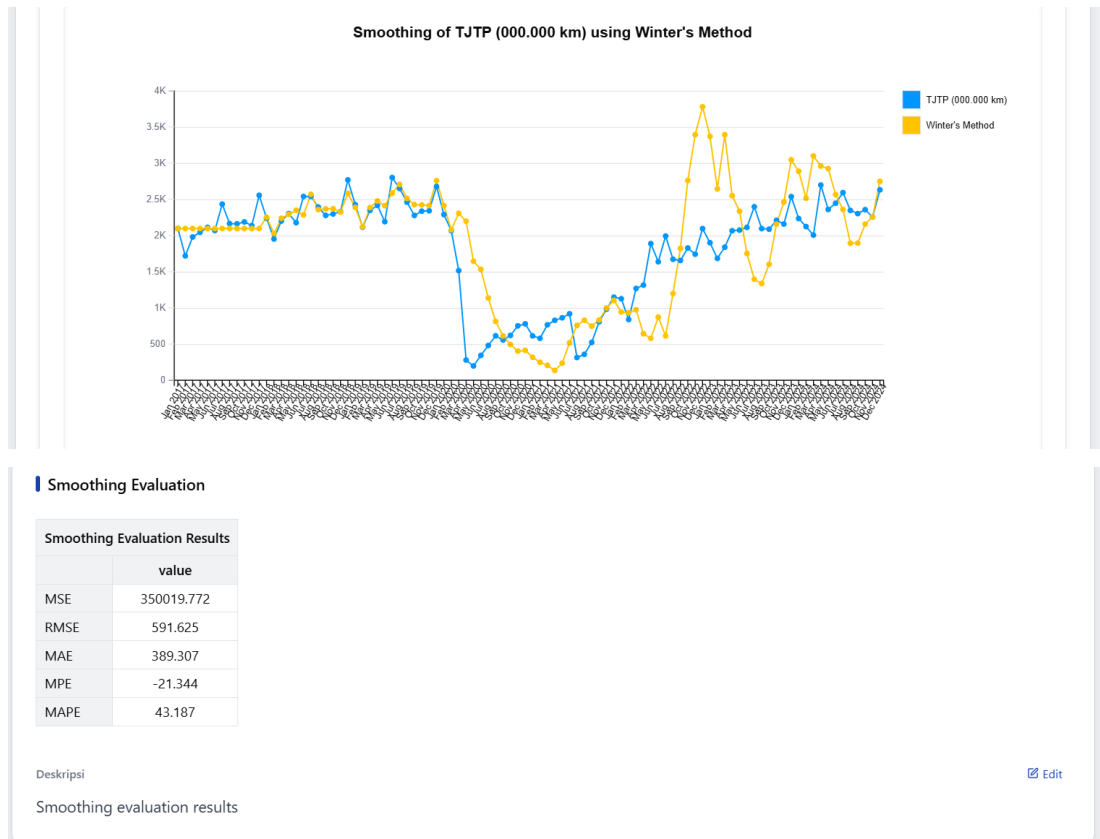
- Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya.

The screenshot shows the 'Smoothing' dialog box with the 'Time' tab selected. The 'Time Option' section shows: 'time spesification : Years-Months', 'year : 2017', 'month : 1', 'periodicity : 12', and 'start time : Jan 2017'. The 'OK' button is highlighted.

- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Smoothing* sebagai berikut.

The screenshot shows the result page titled 'Smoothing Winter's Method Exponential Smoothing'. It contains a 'Description Table' with the following data:

Description Table	
	description
Smoothing Method	Winter's Method
Alpha	0.2
Beta	0.1
Gamma	0.7
Series Name	TJTP (000.000 km)
Series Period	Jan 2017 - Dec 2024
Periodicity	12
Series Length	96
Smoothing Length	96



- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Smoothing* yang disimpan sebagai variabel.

File Edit Data Transform Analyze Graphs Help Statify

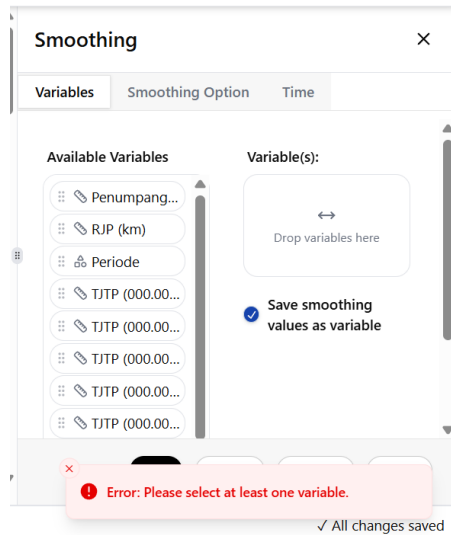
Period	TJTP_000.000_km_sma_5	TJTP_000.000_km_dma_3	TJTP_000.000_km_ses_0.2	TJTP_000.000_km_des_0.4	TJTP_000.000_km_holt_0.1_0.4	TJTP_000.000_km_winter_0.2_0.1_0.7
1 1/01/20...	NaN	NaN	NaN	NaN	2096.00	2096.00
2 1/02/20...	NaN	NaN	2096.00	NaN	2096.00	2096.00
3 1/03/20...	NaN	NaN	2020.40	1692.80	2043.08	2096.00
4 1/04/20...	NaN	NaN	2012.12	1830.08	2018.99	2096.00
5 1/05/20...	1989.80	2208.56	2018.50	1972.66	2004.81	2096.00
6 1/06/20...	1984.20	2201.78	2037.20	2107.94	2003.13	2096.00
7 1/07/20...	2127.20	2397.00	2043.36	2110.35	1999.82	2096.00
8 1/08/20...	2164.20	2331.22	2121.29	2493.78	2050.66	2096.00
9 1/09/20...	2187.60	2305.56	2129.83	2335.16	2074.06	2096.00
10 1/10/20...	2202.60	2082.00	2136.06	2247.47	2098.29	2096.00
11 1/11/20...	2216.40	2095.00	2146.25	2229.80	2126.25	2096.00
12 1/12/20...	2240.80	2462.11	2144.40	2163.26	2146.84	2096.00
13 1/01/20...	2255.60	2420.22	2226.52	2580.97	2223.50	2251.96
14 1/02/20...	2213.60	2176.67	2228.82	2389.05	2261.37	2026.24
15 1/03/20...	2216.00	1929.78	2173.25	1989.66	2254.34	2237.76
16 1/04/20...	2249.20	2100.78	2178.40	2104.23	2270.60	2292.17
17 1/05/20...	2173.60	2341.22	2203.32	2271.57	2296.93	2347.90
18 1/06/20...	2233.80	2541.00	2198.06	2202.44	2303.23	2284.27
19 1/07/20...	2351.60	2599.56	2266.25	2559.62	2354.53	2569.02

Variable Result All changes saved

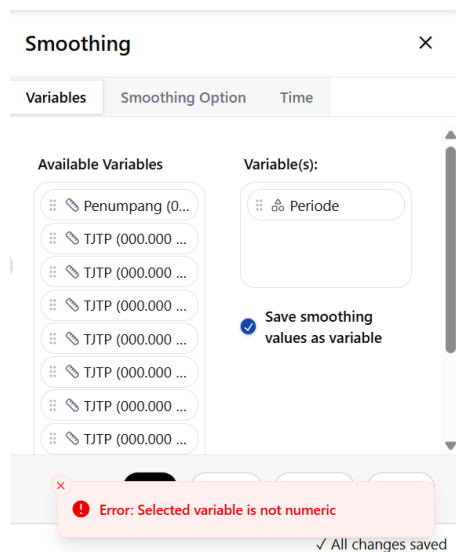
2.7. Validasi Menu

Berikut beberapa validasi dari konfigurasi menu *Smoothing*.

1. Peringatan belum memilih variabel



2. Peringatan ketika menggunakan variabel bukan numerik



3. Peringatan memasukkan parameter *smoothing* lebih dari batasan nilai

Smoothing

Variables Smoothing Option Time

Smoothing Method

method: Winter's Method Exponential Smc

alpha : 2

beta : 0,1

gamma : 0,7

note: winters method need time spesification with

Error: Winter's method alpha parameter must be between 0 and 1.

Smoothing

Variables Smoothing Option Time

Smoothing Method

method: Simple Moving Average

distance : -2

Error: Simple Moving Average period must be between 2 and 11.

✓ All changes saved

4. Peringatan ketika tidak mengonfigurasi periode waktu.

Smoothing

Variables Smoothing Option Time

Time Option

time spesification : Not Dated

periodicity : don't have periodicity

start time : 00:00

Error: Please select another time specification.

✓ All changes saved

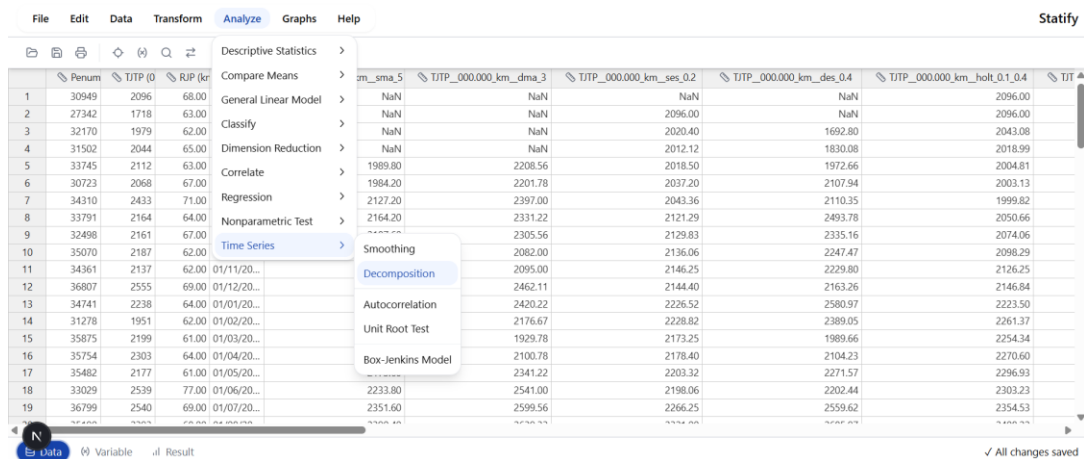
Decomposition

3. Tahapan Menggunakan Menu *Decomposition*

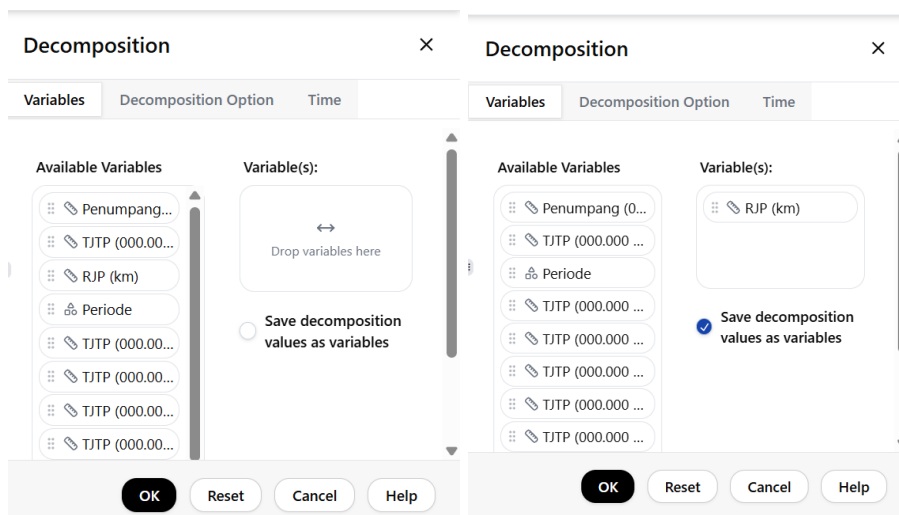
3.1. Tahapan

Ikuti langkah-langkah berikut.

1. Klik Analyze → Time Series → Decomposition untuk masuk ke menu *Decomposition*.



2. Pilih variabel yang ingin digunakan dan klik save decomposition values as variables untuk menyimpan hasil dekomposisi data *time series* sebagai beberapa variabel baru.



3. Pilih tab Decomposition Option untuk memilih metode *decomposition* lalu pilih metode dekomposisi yang diinginkan. Pada menu ini terdapat dua jenis metode dekomposisi yaitu *additive* dan *multiplicative* serta tren linear dan eksponensial yang dapat dipilih jika menggunakan metode multiplikatif. Pada percobaan ini pilih metode

multiplikatif dengan tren linear.

The image shows two side-by-side screenshots of a software interface titled 'Decomposition'. Each window has three tabs: 'Variables', 'Decomposition Option', and 'Time'. The left window is on the 'Decomposition Option' tab, showing a 'Decomposition Method' section with two radio buttons: 'Additive' (selected) and 'Multiplicative'. The right window is also on the 'Decomposition Option' tab, but with 'Multiplicative' selected. It also shows a 'trend' section with two radio buttons: 'Linear' (selected) and 'Exponential'. Both windows have an 'OK' button (highlighted in black) and 'Reset', 'Cancel', and 'Help' buttons at the bottom.

4. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya. Sebagai catatan konfigurasi waktu untuk menu *time series* cukup dilakukan sekali saja karena sudah terintegrasi untuk setiap menu *time series*.

The image shows a screenshot of the 'Decomposition' dialog box with the 'Time' tab selected. The 'Time Option' section contains the following fields: 'time spesification' (a dropdown menu showing 'Years-Months'), 'year' (a text input field with '2017'), 'month' (a text input field with '1'), 'periodicity' (a text input field with '12'), and 'start time' (a text input field with 'Jan 2017'). At the bottom, there are 'OK', 'Reset', 'Cancel', and 'Help' buttons, with 'OK' being highlighted in black.

5. Setelah itu klik ok.
6. Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *decomposition* sebagai berikut.

Decomposition Multiplicative

Description Table

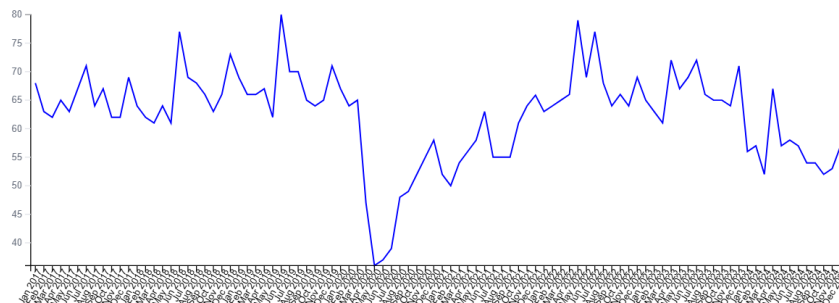
Description Table	
	description
Decomposition Method	Multiplicative Decomposition
Formula of Calculating Decomposition	Classical Decomposition
Trend Method	linear
Series Name	RJP (km)
Series Period	Jan 2017 - Dec 2024
Periodicity	12
Observations	96

Deskripsi

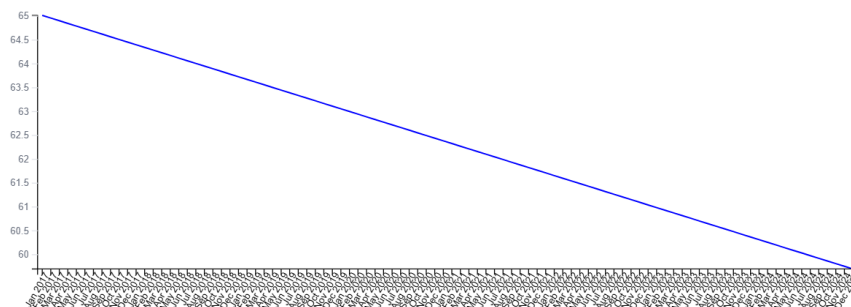
[Edit](#)

Description of the decomposition results

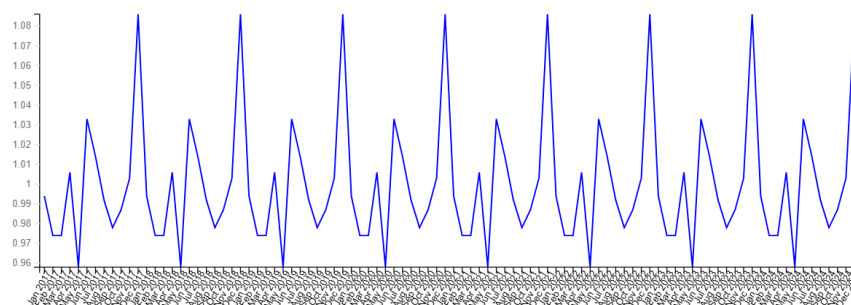
Data Series RJP (km)



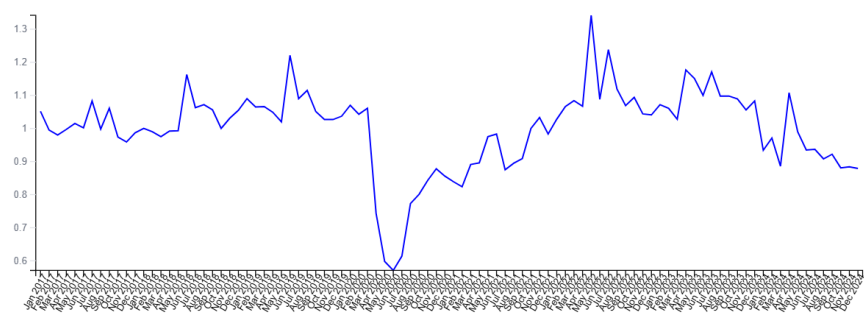
Trend Component of RJP (km)



Seasonal Component of RJP (km)



Irregular Component of RJP (km)



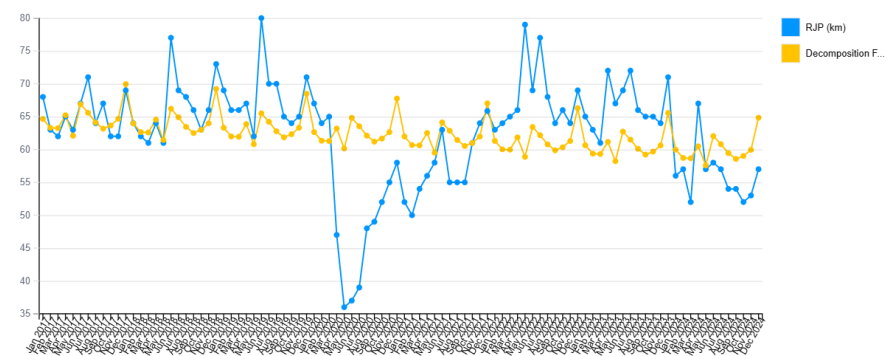
Seasonal Indices

Seasonal Indices Years-Months	
	value
period 1 of 12	0.994
period 2 of 12	0.974
period 3 of 12	0.974
period 4 of 12	1.006
period 5 of 12	0.958
period 6 of 12	1.033
period 7 of 12	1.014
period 8 of 12	0.992
period 9 of 12	0.978
period 10 of 12	0.987
period 11 of 12	1.003
period 12 of 12	1.086

Equation Trend

Linear Trend Equation
$y = 65.074 - 0.056t$

Decomposition Forecasting of RJP (km)



Forecasting Evaluation

Decomposition Forecasting Evaluation Results

	value
MSE	58.802
RMSE	7.668
MAE	5.546
MPE	-1.878
MAPE	9.846

Deskripsi

Evaluation results for the forecasting

[Edit](#)

- Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *Decomposition* yang disimpan sebagai variabel.

Statify											
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📁 📄 🔍 🔄											
	_holt_01_04	TJTP_000.000_km_winter_0.2_0.1_0.7	RIP_km_SC	RIP_km_TC	RIP_km_JC	RIP_km_Forecasting	var	var	var	var	var
1	2096.00	2096.00	0.994	65.018	1.052	64.644					
2	2096.00	2096.00	0.974	64.962	0.995	63.297					
3	2043.08	2096.00	0.974	64.906	0.980	63.245					
4	2018.99	2096.00	1.006	64.850	0.997	65.212					
5	2004.81	2096.00	0.958	64.794	1.015	62.091					
6	2003.13	2096.00	1.033	64.738	1.002	66.894					
7	1999.82	2096.00	1.014	64.682	1.083	65.581					
8	2050.66	2096.00	0.992	64.626	0.998	64.108					
9	2074.06	2096.00	0.978	64.570	1.061	63.163					
10	2098.29	2096.00	0.987	64.514	0.974	63.650					
11	2126.25	2096.00	1.003	64.458	0.959	64.661					
12	2146.84	2096.00	1.086	64.402	0.987	69.944					
13	2223.50	2251.96	0.994	64.347	1.000	63.976					
14	2261.37	2026.24	0.974	64.291	0.990	62.642					
15	2254.34	2237.76	0.974	64.235	0.975	62.591					
16	2270.60	2292.17	1.006	64.179	0.992	64.537					
17	2296.93	2347.90	0.958	64.123	0.993	61.447					
18	2303.23	2284.27	1.033	64.067	1.163	66.200					
19	2354.53	2569.02	1.014	64.011	1.063	64.900					

N
Data

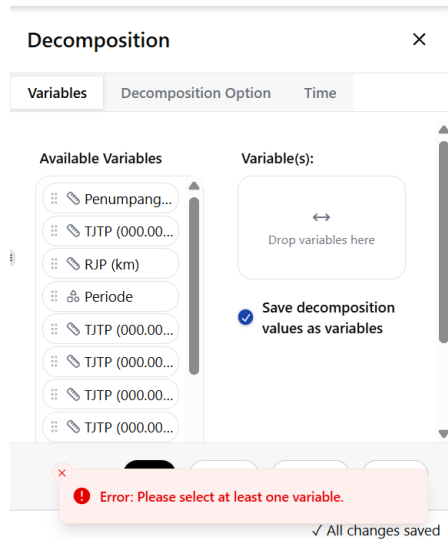
Variable Result

✓ All changes saved

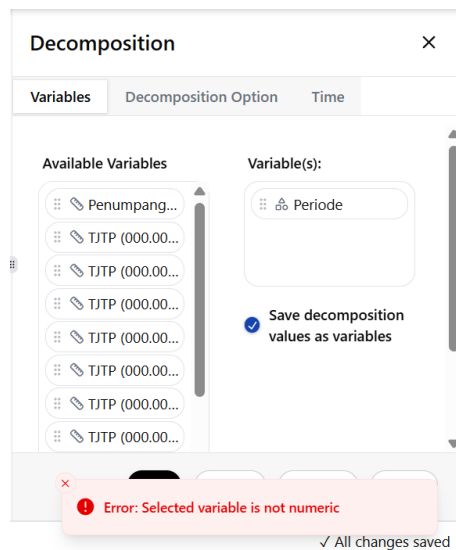
3.2. Validasi menu

Berikut beberapa validasi dari konfigurasi menu *Decomposition*.

1. Peringatan belum memilih variabel



2. Peringatan ketika menggunakan variabel bukan numerik



3. Peringatan ketika memilih periode waktu yang tidak sesuai dengan kelipatan observasi data. Sebagai contoh ketika data dengan periode lima maka jumlah

observasi data harus berjumlah kelipatan lima.

Decomposition

Variables

Decomposition Option

Time

time spesification : Weeks-Work Days(5) ▾

year : 2017

month : 1

day : 31

periodicity : 5

start time : Tuesday, 31 Jan 2017

Error: Data length is not a multiple of the periodicity.

✓ All changes saved

4. Peringatan ketika tidak mengonfigurasi periode waktu.

Decomposition

Variables

Decomposition Option

Time

Time Option

time spesification : Not Dated ▾

periodicity : don't have periodicity

start time : 00:00

Error: Selected time specification doesn't have periodicity.

✓ All changes saved

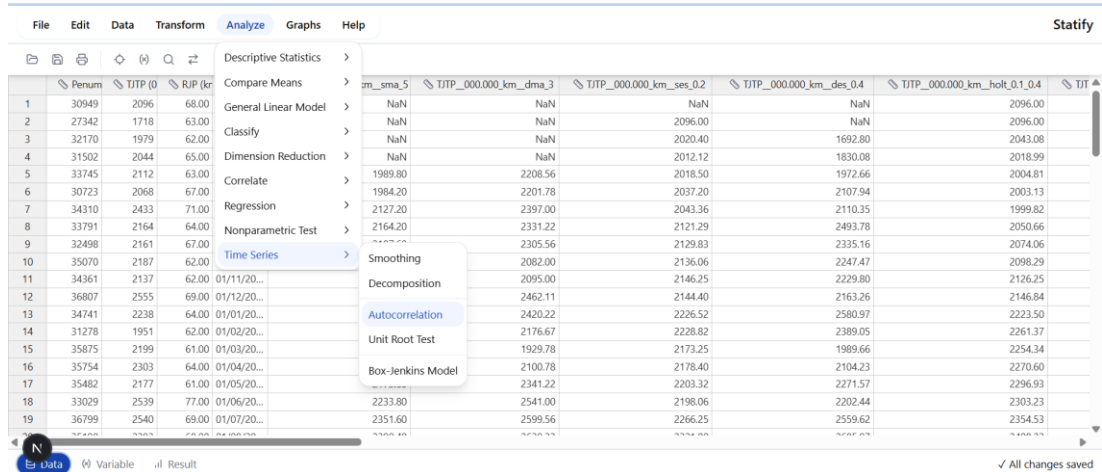
Autocorrelation

4. Tahapan Menggunakan Menu Autocorrelation

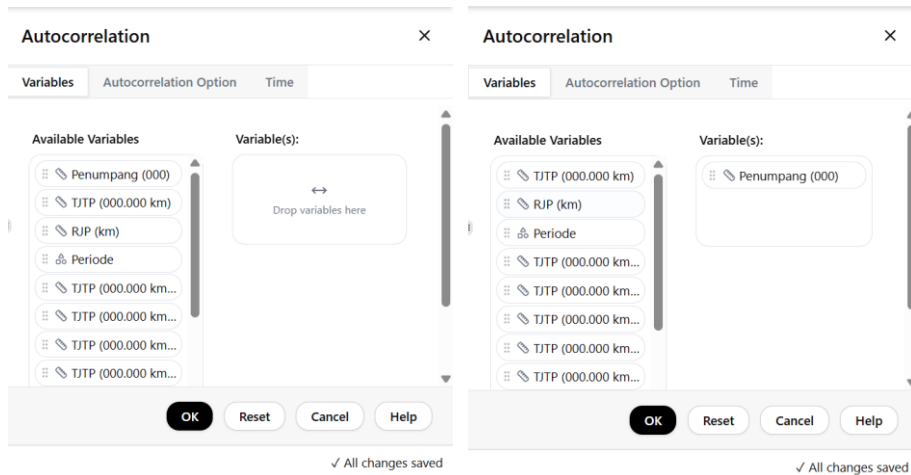
4.1. Tahapan

Ikuti langkah-langkah berikut.

1. Klik Analyze → Time Series → Autocorrelation untuk masuk ke menu *Autocorrelation*.



2. Pilih variabel yang ingin digunakan pada menu *Autocorrelation*.



3. Pilih tab Autocorrelation Option untuk mengatur konfigurasi menu *Autocorrelation*.

The screenshot shows the 'Autocorrelation' dialog box with the 'Autocorrelation Option' tab selected. The 'Variables' tab is also visible. The 'Autocorrelation Options' section contains the following settings: 'maximum lag' is set to 16; 'seasonally difference' is checked; 'autocorrelate on:' has 'first difference' selected (along with 'level' and 'second difference'); and a note states 'Standard error are calculate with Bartlet Approximation'. At the bottom, there are 'OK', 'Reset', 'Cancel', and 'Help' buttons. A status bar at the very bottom indicates '✓ All changes saved'.

4. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya. Sebagai catatan konfigurasi waktu untuk menu *time series* cukup dilakukan sekali saja karena sudah terintegrasi untuk setiap menu *time series*.

The screenshot shows the 'Autocorrelation' dialog box with the 'Time' tab selected. The 'Variables' and 'Autocorrelation Option' tabs are also visible. The 'Time Option' section contains the following settings: 'time spesification' is set to 'Years-Months'; 'year' is set to 2017; 'month' is set to 1; 'periodicity' is set to 12; and 'start time' is set to 'Jan 2017'. At the bottom, there are 'OK', 'Reset', 'Cancel', and 'Help' buttons. A status bar at the very bottom indicates '✓ All changes saved'.

5. Setelah itu klik ok.
6. Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Autocorrelation* sebagai berikut.

Autocorrelation

Description Table

Description Table	
	description
Name Method	Autocorrelation
Series Name	Penumpang (000)
Differencing	first-difference
Seasonal Differencing	with seasonal-differencing
Periodicity	12
Maximum Lag	16
Approach Method of Standard Error	Bartlett Formula
Observations	96
Number Observations of Computable First Lags	82

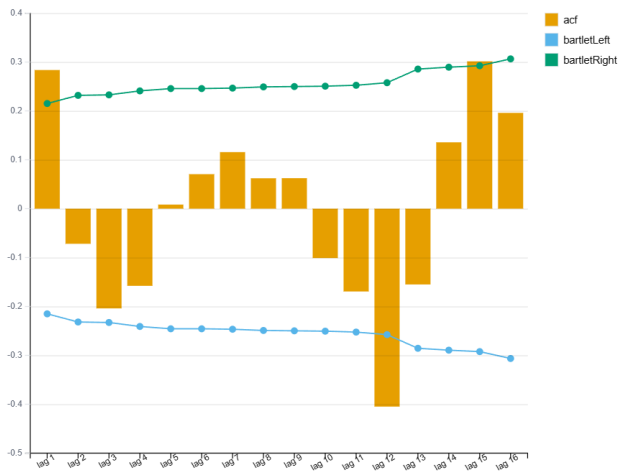
Autocorrelation Table

Autocorrelation Function (ACF)					
	ACF	SE	Ljung-Box	df	p-value
1	0.284	0.110	6.920	1	0.009
2	-0.072	0.118	7.369	2	0.025
3	-0.204	0.119	11.038	3	0.012
4	-0.158	0.123	13.262	4	0.010
5	0.008	0.125	13.268	5	0.021
6	0.071	0.125	13.724	6	0.033
7	0.116	0.126	14.967	7	0.036
8	0.062	0.127	15.330	8	0.053
9	0.062	0.127	15.701	9	0.073
10	-0.101	0.128	16.687	10	0.082
11	-0.169	0.129	19.499	11	0.053
12	-0.405	0.131	35.791	12	0.000
13	-0.155	0.146	38.213	13	0.000
14	0.136	0.148	40.097	14	0.000
15	0.301	0.149	49.521	15	0.000
16	0.196	0.156	53.565	16	0.000

Deskripsi

[Edit](#)

Autocorrelation function results



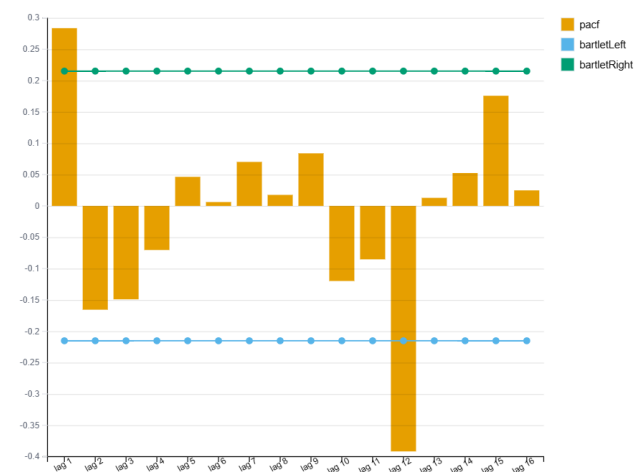
Partial Autocorrelation Table

Partial Autocorrelation Function (PACF)		
	PACF	SE
1	0.284	0.110
2	-0.166	0.110
3	-0.149	0.110
4	-0.070	0.110
5	0.046	0.110
6	0.006	0.110
7	0.070	0.110
8	0.018	0.110
9	0.084	0.110
10	-0.120	0.110
11	-0.085	0.110
12	-0.392	0.110
13	0.013	0.110
14	0.052	0.110
15	0.176	0.110
16	0.025	0.110

Deskripsi

[Edit](#)

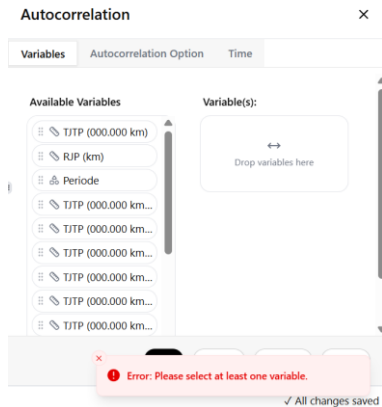
Partial autocorrelation function results



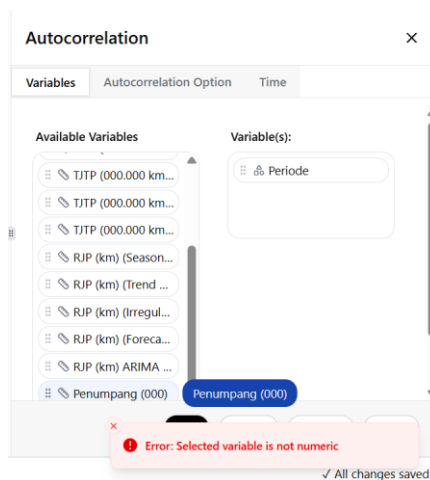
4.2. Validasi Menu

Berikut beberapa validasi dari konfigurasi menu *Autocorrelation*.

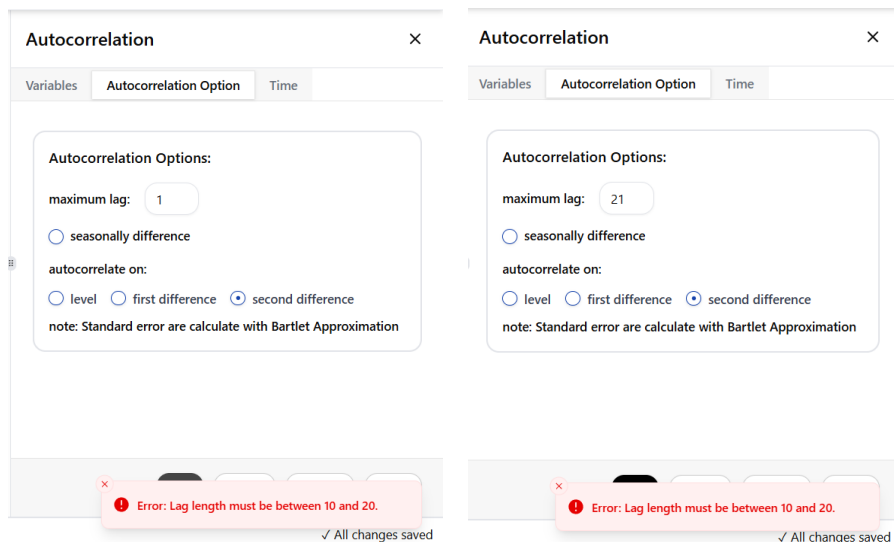
1. Peringatan belum memilih variabel



2. Peringatan ketika menggunakan variabel bukan numerik



3. Peringatan memasukkan jumlah lag maksimal lebih dari batasan nilai



4. Peringatan ketika tidak mengonfigurasi periode waktu ketika mencentang *seasonally difference*.

The image displays two side-by-side screenshots of the 'Autocorrelation' dialog box, illustrating a warning message when the 'seasonally difference' option is selected without proper time configuration.

Left Screenshot (Autocorrelation Option Tab):

- Tab: Autocorrelation Option
- Autocorrelation Options:
 - maximum lag: 18
 - ☒ seasonally difference
 - autocorrelate on:
 - ☐ level
 - ☐ first difference
 - ☒ second difference
 - note: Standard error are calculate with Bartlet Approximation

Right Screenshot (Time Tab):

- Tab: Time
- Time Option:
 - time spesification : Not Dated
 - periodicity : don't have periodicity
 - start time : 00:00

Error Message:

Error: Please select another time specification.

Buttons: OK, Reset, Cancel, Help

Status: ✓ All changes saved

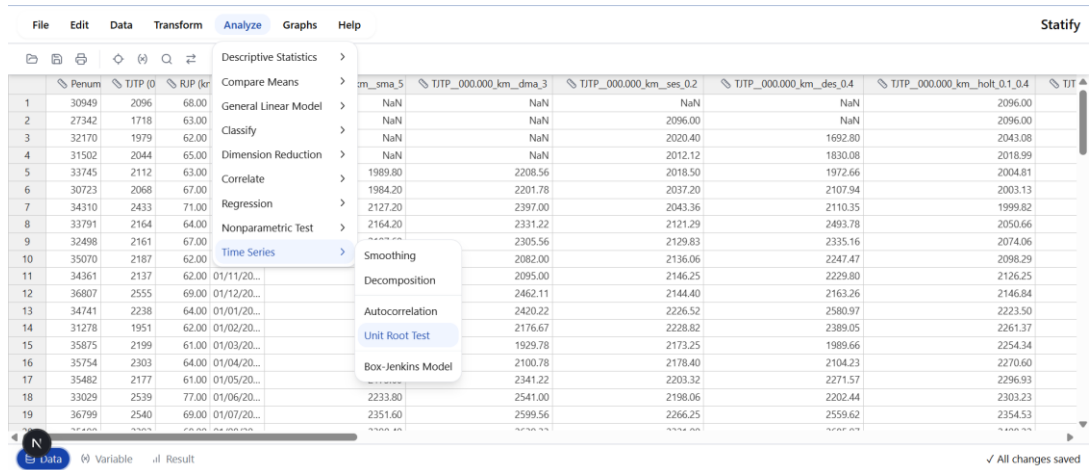
Unit Root Test

5. Tahapan Menggunakan Menu *Unit Root Test*

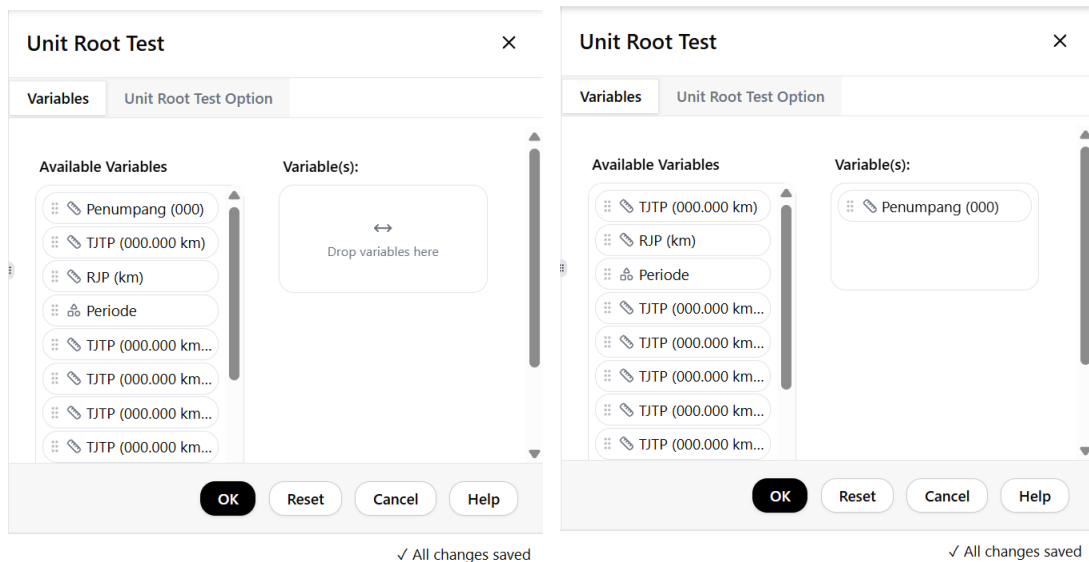
5.1. *Dickey Fuller-Test*

Ikuti langkah-langkah berikut.

1. Klik Analyze → Time Series → Unit Root Test untuk masuk ke menu *Unit Root Test*.



2. Pilih variabel yang ingin digunakan pada menu *Unit Root Test*.



- Pilih tab Unit Root Test Option untuk mengatur konfigurasi menu *Unit Root Test*.
Pilih metode *Dickey-Fuller Test* untuk pengujian awal

Unit Root Test

Variables Unit Root Test Option

Method:

☒ dickey-fuller
☐ augmented dickey-fuller

Unit Root Test Options:

calculate on:

☐ level ☒ first difference ☐ second difference

equation:

☐ none ☒ intercept ☐ trend and intercept

OK Reset Cancel Help

✓ All changes saved

- Setelah itu klik ok.
- Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil penghitungan *Unit Root Test* sebagai berikut.

Unit Root Test: Dickey-Fuller

Description Table

	description
Name Method	Dickey-Fuller
Series Name	Penumpang (000)
Equation	no_trend
Number of Lags	0
Differencing	first-difference
Probability Value	Use MacKinnon (1996) one-sided p-values
Observations	96
Number Observations After Computing	94

Dickey-Fuller Test Statistic

Dickey-Fuller Test Statistic		
	t-statistic	p-value
Dickey-Fuller Test	-9.469	0.000
Critical Value	1%	-3.501
	5%	-2.892
	10%	-2.583

Deskripsi

Unit root test statistic results

Edit

Coefficient Regression Test

Coefficients Test				
	coefficient	standard error	t-statistic	p-value
Constant	194.777	337.084	0.578	0.565
Penumpang (000)	-0.985	0.104	-9.469	0.000

Deskripsi

[Edit](#)

Coefficients of the regression used in the unit root test

Selection Criterion

Selection Criterion	
	value
R-Squared	0.494
Adj. R-Squared	0.488
S.E. of Regression	3265.761
Sum Squared Resid	981198108.482
Log Likelihood	-892.947
F-Statistic	89.658
Prob(F-Stat)	0.000
Mean Dependent Var	72.830
S.D. Dependent Var	4564.261
Akaike Info Crit	19.041
Schwarz Criterion	19.096
Hannan-Quinn	19.063
Durbin-Watson	1.931

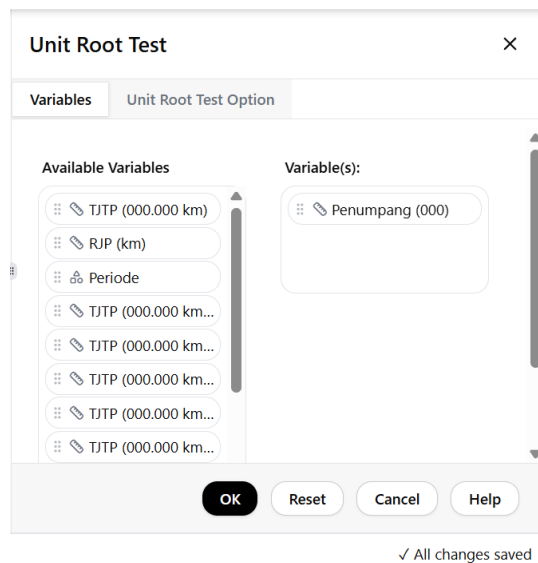
5.2. Augmented Dickey Fuller-Test

Ikuti langkah-langkah berikut.

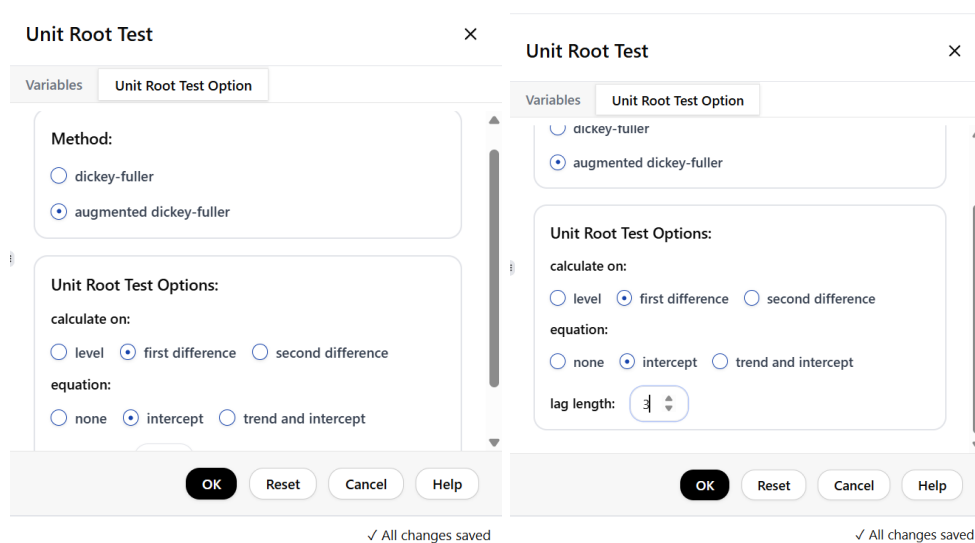
1. Klik Analyze → Time Series → Unit Root Test untuk masuk ke menu *Unit Root Test*.

The screenshot shows the EViews software interface. The 'Analyze' menu is open, and the path 'Time Series' → 'Unit Root Test' is highlighted. The background displays a data table with columns for 'Penumpang' (Passengers) and 'TJTP' (Total Journey Time) across various dates. The 'Unit Root Test' option is selected in the 'Time Series' submenu.

2. Konfigurasi variabel tidak perlu dilakukan karena aplikasi menyimpan hasil konfigurasi sebelumnya.



3. Pilih tab Unit Root Test Option untuk mengatur konfigurasi menu *Unit Root Test*. Pilih metode *Augmented Dickey-Fuller Test* untuk pengujian lanjutan. Pada metode ini terdapat tambahan masukan parameter yaitu jumlah lag.



4. Setelah itu klik ok.
5. Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil penghitungan *Unit Root Test* sebagai berikut.

Unit Root Test: Augmented Dickey-Fuller

Description Table

Description Table	
	description
Name Method	Augmented Dickey-Fuller
Series Name	Penumpang (000)
Equation	no_trend
Number of Lags	3
Differencing	first-difference
Probability Value	Use MacKinnon (1996) one-sided p-values
Observations	96
Number Observations After Computing	91

Augmented Dickey-Fuller Test Statistic

Augmented Dickey-Fuller Test Statistic			
		t-statistic	p-value
Augmented Dickey-Fuller Test		-5.173	0.000
Critical Value	1%	-3.504	
	5%	-2.894	
	10%	-2.584	

Deskripsi

[Edit](#)

Unit root test statistic results

Coefficient Regression Test

Coefficients Test				
	coefficient	standard error	t-statistic	p-value
Constant	146.600	348.780	0.420	0.675
Penumpang (000) Diff Lag (1)	0.129	0.181	0.710	0.480
Penumpang (000) Diff Lag (2)	0.112	0.149	0.753	0.453
Penumpang (000) Diff Lag (3)	0.086	0.107	0.807	0.422
Penumpang (000)	-1.091	0.211	-5.173	0.000

Deskripsi

[Edit](#)

Coefficients of the regression used in the unit root test

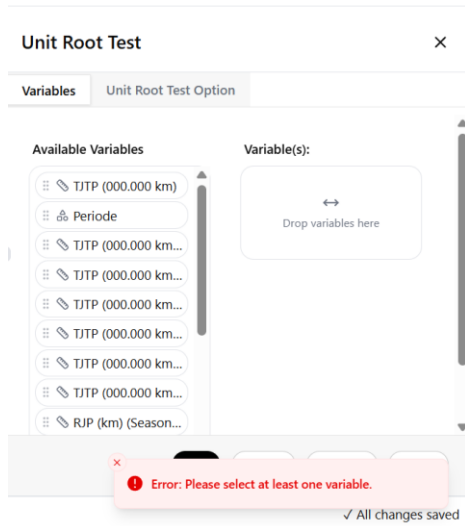
Selection Criterion

Selection Criterion	
	value
R-Squared	0.483
Adj. R-Squared	0.459
S.E. of Regression	3314.716
Sum Squared Resid	944911531.343
Log Likelihood	-864.210
F-Statistic	20.095
Prob(F-Stat)	0.000
Mean Dependent Var	10.945
S.D. Dependent Var	4506.883
Akaike Info Crit	19.104
Schwarz Criterion	19.241
Hannan-Quinn	19.159
Durbin-Watson	1.955

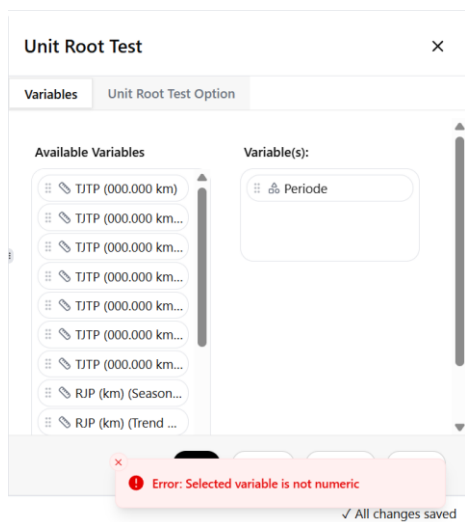
5.3. Validasi Menu

Berikut beberapa validasi dari konfigurasi menu *Unit Root Test*.

1. Peringatan belum memilih variabel



2. Peringatan ketika menggunakan variabel bukan numerik



3. Peringatan memasukkan *lag length* metode *augmented dickey-fuller* lebih dari batasan nilai

The image displays two side-by-side screenshots of the 'Unit Root Test' dialog box, illustrating an error message when the lag length is set outside the valid range of 1 to 10.

Left Screenshot:

- Method:** ☒ augmented dickey-fuller
- Unit Root Test Options:**
 - calculate on:** ☐ level ☐ first difference ☒ second difference
 - equation:** ☐ none ☒ intercept ☐ trend and intercept
 - lag length:** -1

Right Screenshot:

- Method:** ☒ augmented dickey-fuller
- Unit Root Test Options:**
 - calculate on:** ☐ level ☐ first difference ☒ second difference
 - equation:** ☐ none ☒ intercept ☐ trend and intercept
 - lag length:** 12

Error Message (Bottom of both):

❌ Error: Lag length must be between 1 and 10.

✓ All changes saved

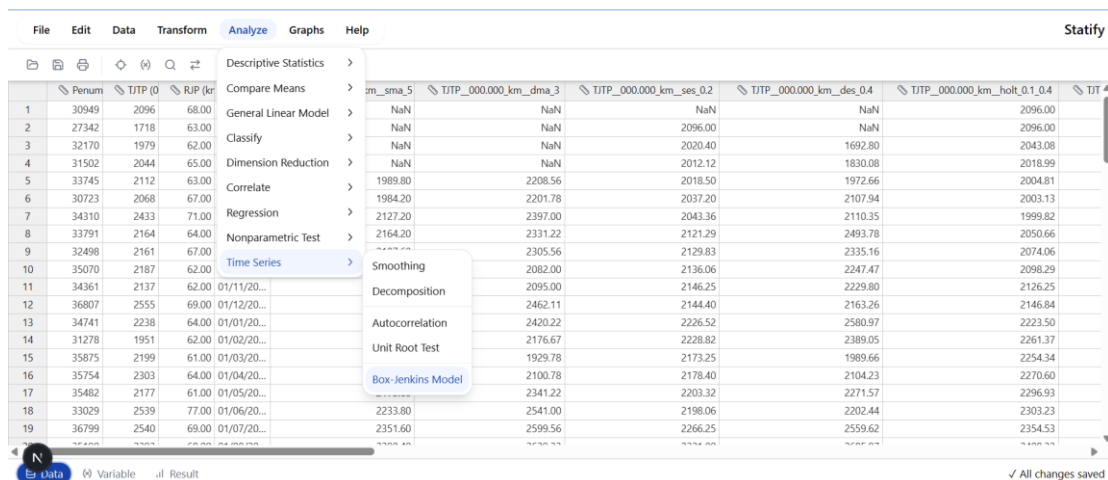
Box-Jenkins Model

6. Tahapan Menggunakan Menu Box-Jenkins Model

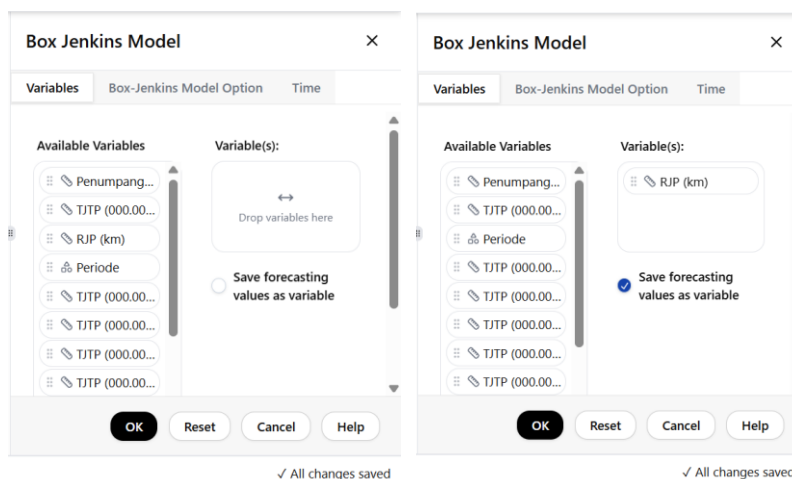
6.1. Tahapan

Ikuti langkah-langkah berikut.

6. Klik Analyze → Time Series → Box-Jenkins Model untuk masuk ke menu *Box-Jenkins Model*.

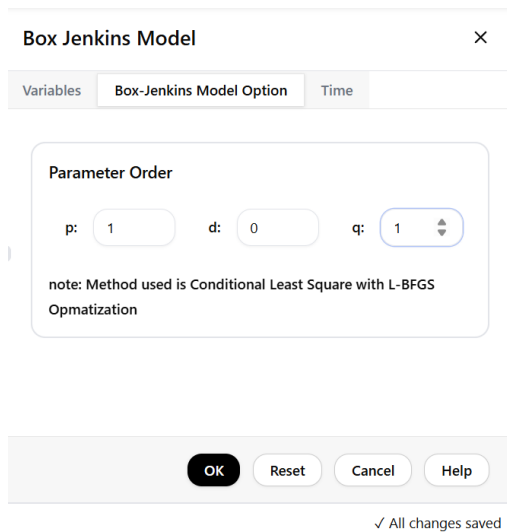


7. Pilih variabel yang ingin digunakan dan klik save forecasting values as variable untuk menyimpan hasil peramalan *box-jenkins model* (ARIMA) sebagai beberapa variabel baru.



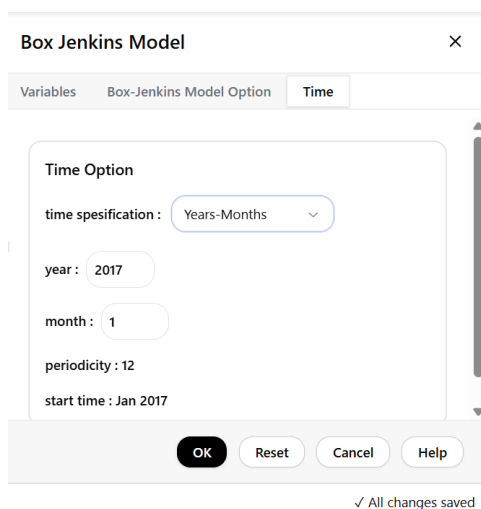
8. Pilih tab Box-Jenkins Model Option untuk memilih inputan parameter model, yakni parameter p menunjukkan jumlah parameter *auto-regressive*, parameter d menunjukkan jumlah differensiasi model, dan parameter q menunjukkan jumlah parameter *moving*

average.



The image shows a dialog box titled "Box Jenkins Model" with a close button (X). It has three tabs: "Variables", "Box-Jenkins Model Option", and "Time". The "Box-Jenkins Model Option" tab is selected. Inside the dialog, there is a section titled "Parameter Order" with three input fields: "p:" with value 1, "d:" with value 0, and "q:" with value 1. Below these fields, a note states: "note: Method used is Conditional Least Square with L-BFGS Opmatization". At the bottom of the dialog, there are four buttons: "OK", "Reset", "Cancel", and "Help". Below the dialog, a status bar indicates "✓ All changes saved".

9. Pilih tab time untuk menyesuaikan periode waktu, namun dikarenakan konfigurasi sudah dilakukan di tahapan sebelumnya periode waktu pada tab ini dapat diabaikan karena aplikasi menyimpan hasil konfigurasi sebelumnya. Sebagai catatan konfigurasi waktu untuk menu *time series* cukup dilakukan sekali saja karena sudah terintegrasi untuk setiap menu *time series*.



The image shows the same "Box Jenkins Model" dialog box, but with the "Time" tab selected. The "Time Option" section contains a "time spesification" dropdown menu set to "Years-Months". Below this are input fields for "year" (2017), "month" (1), and "periodicity" (12). The "start time" is displayed as "Jan 2017". At the bottom, there are "OK", "Reset", "Cancel", and "Help" buttons. A status bar at the bottom indicates "✓ All changes saved".

10. Setelah itu klik ok.
11. Selanjutnya Anda akan dipindahkan ke halaman result dan Anda dapat melihat hasil *Box-Jenkins Model* sebagai berikut.

ARIMA(1,0,1)

Description Table

Description Table	
	description
Name Method	ARIMA (1, 0, 1)
Estimation Method	Conditional Least Squares (CLS)
Function Estimation	Conditional Sum of Squares (CSS)
Optimization Method	L-BFGS
Series Name	RJP (km)
Series Period	Jan 2017 - Dec 2024
Series Length	96

Deskripsi

Edit

Description of the ARIMA results

Coefficient Test for ARIMA(1,0,1)

Coefficients Test for ARIMA (1,0,1)				
	coef	std. error	t value	p-value
Constant	62.841	1.582	39.732	0.000
AR(1)	0.839	0.061	13.682	0.000
MA(1)	0.230	0.117	1.959	0.053

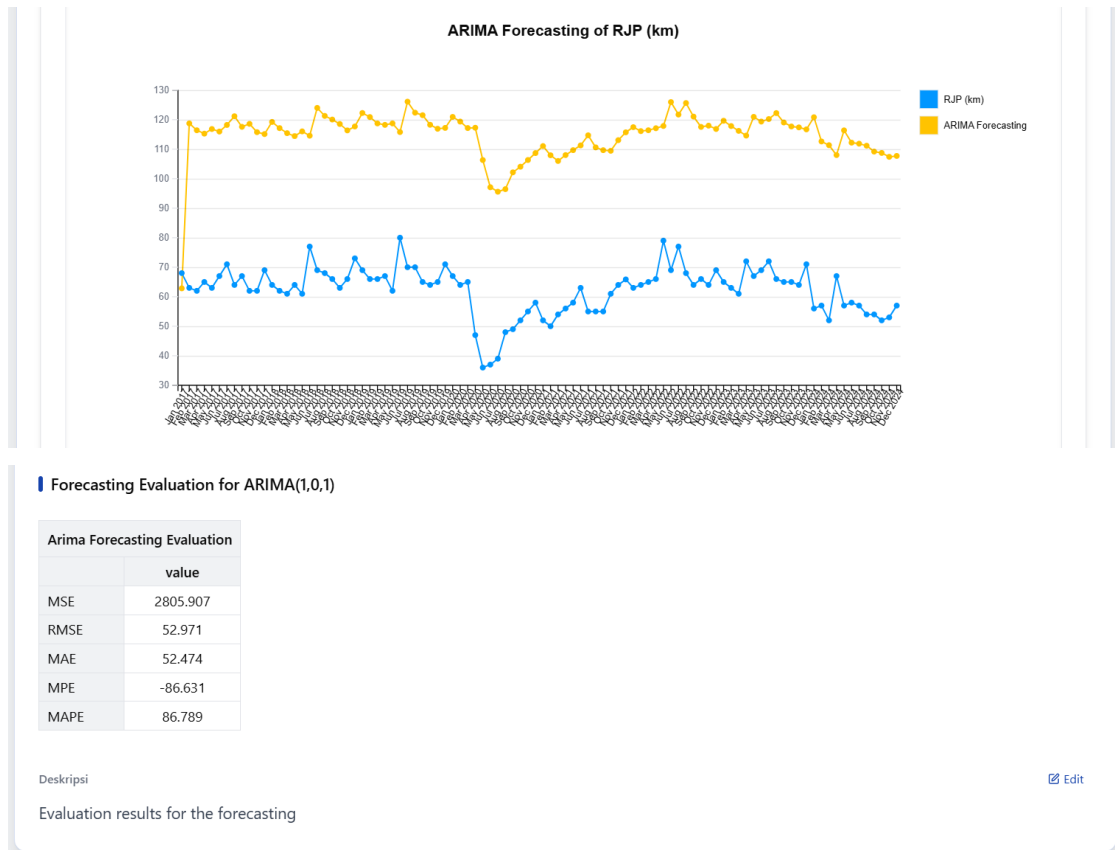
Deskripsi

Edit

Coefficient test results

Criteria Selection for ARIMA(1,0,1)

Selection Criteria for RJP (km)	
	value
R-Squared	0.548
Adj. R-Squared	0.539
S.E. of Regression	5.486
Sum Squared Resid	2799.363
Log Likelihood	-298.112
F-Statistic	112.877
Prob(F-Stat)	0.000
Mean Dependent Var	62.353
S.D. Dependent Var	8.077
Akaike Info Crit	6.273
Schwarz Criterion	6.353
Hannan-Quinn	6.306
Durbin-Watson	2.031



12. Klik tombol data yang terdapat pada sisi kiri bawah halaman untuk melihat hasil *forecasting Box-Jenkins Model* yang disimpan sebagai variabel.

File Edit Data Transform Analyze Graphs Help										Statify	
TJTP_000.000_km_holt_0.1_0.4 TJTP_000.000_km_winter_0.2_0.1_0.7 RJP_km_SC RJP_km_TC RJP_km_IC RJP_km_Forecasting RJP_km_ARIMA_1_0_1 var var											
1	2096.00	2096.00	0.994	65.018	1.052	64.644	62.841				
2	2096.00	2096.00	0.974	64.962	0.995	63.297	118.727				
3	2043.08	2096.00	0.974	64.906	0.980	63.245	116.402				
4	2018.99	2096.00	1.006	64.850	0.997	65.212	115.258				
5	2004.81	2096.00	0.958	64.794	1.015	62.091	116.824				
6	2003.13	2096.00	1.033	64.738	1.002	66.894	115.964				
7	1999.82	2096.00	1.014	64.682	1.083	65.581	118.205				
8	2050.66	2096.00	0.992	64.626	0.998	64.108	121.158				
9	2074.06	2096.00	0.978	64.570	1.061	63.163	117.570				
10	2098.29	2096.00	0.987	64.514	0.974	63.650	118.574				
11	2126.25	2096.00	1.003	64.458	0.959	64.661	115.757				
12	2146.84	2096.00	1.086	64.402	0.987	69.944	115.110				
13	2223.50	2251.96	0.994	64.347	1.000	63.976	119.228				
14	2261.37	2026.24	0.974	64.291	0.990	62.642	117.126				
15	2254.34	2237.76	0.974	64.235	0.975	62.591	115.425				
16	2270.60	2292.17	1.006	64.179	0.992	64.537	114.424				
17	2296.93	2347.90	0.958	64.123	0.993	61.447	116.023				
18	2303.23	2284.27	1.033	64.067	1.163	66.200	114.561				
19	2354.53	2569.02	1.014	64.011	1.063	64.900	123.978				

Variable Result

All changes saved

6.2. Validasi Menu

Berikut beberapa validasi dari konfigurasi menu *Box-Jenkins Model*.

1. Peringatan belum memilih variabel

The screenshot shows the 'Box Jenkins Model' configuration window with the 'Variables' tab selected. The 'Available Variables' list on the left contains several items, including 'Penumpang (000)', 'TJTP (000.000 km)', and 'Periode'. The 'Variable(s):' box on the right is empty, with a 'Drop variables here' instruction. A red error message at the bottom states: 'Error: Please select at least one variable.' Below the error message, it says 'All changes saved'.

2. Peringatan ketika menggunakan variabel bukan numerik

The screenshot shows the 'Box Jenkins Model' configuration window with the 'Variables' tab selected. The 'Available Variables' list on the left contains several items, including 'Penumpang (000)', 'TJTP (000.000 km)', and 'RJP (km) (Season...)'. The 'Variable(s):' box on the right contains 'Periode'. A red error message at the bottom states: 'Error: Selected variable is not numeric.' Below the error message, it says 'All changes saved'.

3. Peringatan memasukkan parameter model lebih dari batasan nilai

The screenshot shows the 'Box Jenkins Model' configuration window with the 'Box-Jenkins Model Option' tab selected. The 'Parameter Order' section shows 'p: -1', 'd: 0', and 'q: 0'. A red error message at the bottom states: 'Error: AR order must be between 0 and 5.' Below the error message, it says 'All changes saved'.

The screenshot shows the 'Box Jenkins Model' configuration window with the 'Box-Jenkins Model Option' tab selected. The 'Parameter Order' section shows 'p: 1', 'd: 3', and 'q: 0'. A red error message at the bottom states: 'Error: Differencing order must be between 0 and 2.' Below the error message, it says 'All changes saved'.

Box Jenkins Model

Variables

Box-Jenkins Model Option

Time

Parameter Order

p: 1 d: 2 q: 9

note: Method used is Conditional Least Square with L-BFGS Opmatization

Error: MA order must be between 0 and 5.

✓ All changes saved

4. Peringatan ketika tidak mengonfigurasi periode waktu.

Box Jenkins Model

Variables

Box-Jenkins Model Option

Time

Time Option

time spesification : Not Dated

periodicity : don't have periodicity

start time : 00:00

Error: Please select another time specification.

✓ All changes saved